# Detection, Adaptation, and Watermark Persistence
# A Pedagogical Research Report

## Summary of contributions

This report evaluates AI-content detection, adaptive generation and watermark removal through reproducible text and image experiments. Its contributions are the controlled comparisons, paired output evidence, preservation-aware accounting and documented negative results.

**Detector generalization.** Ten text classifiers and five image classifiers are compared across datasets. Qwen reaches macro F1 of 0.9236 on MAGE and 0.8310 on Pile; Gemma reaches 0.9499 on peer reviews. Image transfer is weaker: EfficientNet falls from 0.9346 on ArtiFact to 0.3333 on CIFAKE. A ConvNeXt augmentation intervention fails its acceptance criterion. These measurements show how dataset and processing shifts affect the retained classifiers.

**Adaptive generators and detector retraining.** GRPO trains Qwen3.5-9B-Base LoRA attackers against Gemma, ModernBERT-large and a seven-detector ensemble. One defender update reduces evasion on identical saved outputs from 77.2% to 0.2%, 96.8% to 1.0%, and 94.8% to 0.4%, respectively. Each branch evaluates 500 generated and 500 human passages; attacker budgets differ. For SD1.5 against DDA, a validation-selected DDPO checkpoint raises held-out evasion from 1.2% to 8.2% while retaining 99.1% of mean CLIP alignment. A later 99.4%-evasion checkpoint retains only 63.3%, exposing the utility cost.

**Watermark removal and content preservation.** The experiments cover TextSeal 2 and a local distilled text watermark on Qwen3.5-4B instruct, plus released DistSeal RAR-XL and UViT-H/DC-AE image generators. Rewriting, probability access, weight changes and output purification have distinct access contracts. Raw detection losses are joined with paired-content checks. Under the retained three-metric image rule, 2,500 clean LoRA updates produce 339/512 joint removals for DC-AE and 0/512 for RAR. RAR still retains class alignment on many outputs; changed sampled scenes fail the paired-image rule. Examples and component counts make that distinction visible.

**Robustness candidates and probability interfaces.** Image meta-distillation yields no reliable defense gain: the revised post-hoc analysis gives 29/64 joint removals for the released model and 28/64 for each variant. On 128 fresh TextSeal 2 questions, conditional probability exposure reduces joint direct-resampling removals from 114/128 to 19/128, but an adaptive ratio client removes 95 of 96 quality-passing outputs. Support pruning reduces ratio-client joint removals from 21/32 to 3/32 or 1/32 on reused development sources. Fresh confirmation of pruning remains outstanding.

**Calibration and reproducibility.** An independent human-null diagnostic flags 5/4,096 prefixes at the unchanged nominal TextSeal 2 threshold of 0.001. Frozen inputs, source hashes, saved outputs and explicit selection rules support the comparisons. Single-key results, automatic preservation proxies, post-hoc reanalyses and the absence of a second adaptive attacker round bound the conclusions.

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# 1 Research questions and reading guide

This project asks how AI-content detectors generalize, what happens when generators adapt to them, and how long deliberate watermark signals survive. Detector scores measure statistical evidence. A watermark differs from a content classifier because a provider deliberately inserts the signal that its detector seeks.

Section 2 benchmarks text and image detectors. Section 3 uses GRPO and DDPO to train generators against them. Section 4 defines watermark detection and preservation, then compares removal methods. Sections 5 and 6 test image meta-distillation and TextSeal 2 probability interfaces.

Read each result alongside the attacker's access, the held-out cohort, baseline detection, and the preservation rule. For example, fewer successful rewrites can mean either that the watermark survived or that the new answers failed the content checks. The tables report those outcomes separately.

**Notation used in the report.** Each mathematical section states its own objects and dimensions. The sets $\mathbb{R}$, $\mathbb{N}_{>0}$ and $\mathbb{N}_0$ contain real numbers, positive integers and nonnegative integers. For $d, m, n \in \mathbb{N}_{>0}$, a scalar has one value, $a \in \mathbb{R}^d$ is a vector with $d$ coordinates, and $A \in \mathbb{R}^{m \times n}$ is an $m$-row, $n$-column matrix. For a finite set $\mathcal{V}$ of size $V \in \mathbb{N}_{>0}$, its probability simplex is

$$\Delta(\mathcal{V}) = \left\{ p = (p_v)_{v \in \mathcal{V}} \in [0,1]^V : \sum_{v \in \mathcal{V}} p_v = 1 \right\}. \tag{1}$$

The set $\mathcal{V}^*$ contains all finite token sequences, including the empty sequence. The notation $\mathbf{1}\{E\} \in \{0,1\}$ indicates whether statement $E$ holds. A superscript $\top$ denotes transpose, $\|\cdot\|_2$ is the Euclidean norm, and $\mathbb{E}$ denotes expectation under the stated distribution. For parameters $\theta \in \mathbb{R}^{d_\theta}$ with $d_\theta \in \mathbb{N}_{>0}$, the gradient of a differentiable scalar function $f : \mathbb{R}^{d_\theta} \to \mathbb{R}$ has dimension $\nabla_\theta f \in \mathbb{R}^{d_\theta}$, with one entry per parameter. Logarithms are natural. For discrete distributions $p, q \in \Delta(\mathcal{V})$, their scalar Kullback–Leibler divergence is

$$\mathrm{KL}(p\|q) = \sum_{v:p_v>0} p_v \log(p_v/q_v) \in [0,\infty], \tag{2}$$

with infinite divergence if $q_v = 0 < p_v$. The continuous analogue integrates the density-weighted log ratio. Numerical parameters in displayed examples illustrate a mechanism; experiment tables give the measured outcomes.

# 2 Supervised detection and generalization

## 2.1 Text detection across datasets

**Question.** Before adapting a generator, the project first asks which supervised detectors transfer beyond their fitting distribution. This subsection requires only the supervised-learning distinction between training, validation and test sets. Its results are independent of the subsequent reinforcement-learning and watermark experiments.

**Models and evidence.** The retained MAGE notebook compares four TF-IDF classifiers (logistic regression, random forest, XGBoost and LightGBM) with six QLoRA classifiers: ModernBERT-base, ModernBERT-large, Qwen3.5-4B-Base, Gemma-4-E4B, Nanbeige4.2-3B and LFM2.5-2.6B. Text labels in this notebook are machine $= 0$ and human $= 1$. The benchmark reports accuracy, macro F1, ROC-AUC and PR-AUC. Macro F1 gives equal weight to the two class-specific F1 values; it is not automatically the F1 of the AI class alone.

| Model | MAGE F1 | Pile F1 | Peer-review F1 |
|---|---|---|---|
| TF-IDF logistic regression | .6588 | .6670 | .6236 |
| TF-IDF random forest | .6687 | .6321 | .6105 |
| TF-IDF XGBoost | .6836 | .6109 | .6259 |
| TF-IDF LightGBM | .6994 | .6228 | .6757 |
| ModernBERT-base | .7908 | .7282 | .6249 |
| ModernBERT-large | .8153 | .7473 | .7700 |
| Qwen3.5-4B-Base | .9236 | .8310 | .8093 |
| Gemma-4-E4B | .9075 | .8006 | .9499 |
| Nanbeige4.2-3B | .7004 | .7272 | .6223 |
| LFM2.5-2.6B | .9061 | .7633 | .9154 |

Table 1: Supervised AI-text detection: macro F1 of ten detectors fitted on MAGE, from retained notebook job 1145971. Columns evaluate the same fitted model on the MAGE holdout, external Pile texts and 2,000 peer reviews (1,000 human and 1,000 generated across five LLMs). Rows name the detector architecture; transformer rows use QLoRA classification. Higher macro F1 is better. These are single-run distribution comparisons, not separate retraining per column.

**Reading the result.** Qwen has the highest observed MAGE and Pile macro F1, while Gemma has the highest peer-review macro F1. A model ranking on one dataset therefore does not define a universal detector ranking. An encoder can be cheaper to use than a decoder classifier, but this experiment does not provide a matched-compute efficiency frontier.

The peer-review evaluation contains 2,000 items: 1,000 human reviews and 200 generated reviews from each of GPT-4o, Gemini-1.5-Pro, Claude-Sonnet-3.5, Llama-3.1-70B and Qwen-2.5-72B. Its text extraction omits metadata fields such as review ratings and identifiers. This reduces an obvious metadata shortcut, but the evaluation changes both domain and generator. It cannot identify a purely temporal generalization effect. The executed MAGE dataset did not expose the additional MAGE-specific OOD splits named in the notebook introduction; those absent split evaluations must not be reported as completed.

**Why this matters later.** The defender-comparison workflow selects the strongest retained non-Qwen decoder and the strongest encoder from the MAGE comparison, then warms them against Qwen-generated continuations. Because selection uses the MAGE test ranking, that ranking is a selection statistic for the later study, not a second untouched confirmation. The CNN/DailyMail prompt evaluation remains a distinct experiment with its own split and overlap audit.

**Limits.** These are single-run observations without uncertainty intervals. High classifier scores are not calibrated authorship probabilities. Token limits, class order and dataset-specific style can alter apparent performance. Neither these scores nor later evasion results certify the truth of a document's content.

## 2.2 Image detection across datasets

**Question.** Can a classifier trained on a broad collection of real and generated images recognize images from a different acquisition pipeline? This subsection is independent of DDPO and watermarking: it concerns supervised classifiers and their generalization.

The image notebook starts from ArtiFact's 2,440,031 available images, takes a seeded stratified 15% sample, and reserves a stratified 20% holdout from that sample. This is an image-level holdout, not a guarantee of unseen-generator evaluation. Five models are compared: EfficientNet-B4, a CLIP classifier, ConvNeXt V2 Tiny, DINOv2 ViT-base and ResNet-50. CIFAKE and Shoes serve as external evaluations. The CIFAKE input resolution differs sharply from the ArtiFact pipeline, making resolution and processing part of the shift.

| Model | ArtiFact F1 | CIFAKE F1 | Shoes F1 |
|---|---|---|---|
| EfficientNet-B4 | .9346 | .3333 | .6700 |
| CLIP classifier | .8039 | .3377 | .6732 |
| ConvNeXt V2 Tiny | .3811 | .3333 | .3834 |
| DINOv2 ViT-base | .6245 | .4564 | .3233 |
| ResNet-50 | .7562 | .3786 | .5398 |

Table 2: Supervised AI-image detection: macro F1 of five classifiers fitted on the seeded 15% ArtiFact sample, from notebook job 1153220. Columns evaluate each fitted classifier on its stratified ArtiFact holdout, external CIFAKE and external Shoes. Rows name the classifier backbone. Higher macro F1 is better; values near one third on balanced data can reflect single-class collapse. No generator adaptation or watermark removal enters this table.

For example, a classifier predicting one class on a balanced dataset has 50% accuracy but macro F1 of one third: the predicted class has precision $1/2$ and recall one, hence F1 $2/3$, while the other class has F1 zero. Averaging yields $1/3$. Thus the CIFAKE rows near 0.3333 are consistent with unhelpful single-class predictions, not moderately successful discrimination.

**The intervention and its rejection.** The project retrains ConvNeXt on balanced ArtiFact data with downsampling/upsampling, JPEG recompression, blur, color changes and flips. The prespecified ex-

ploratory gate requires macro-F1 improvement on both external datasets and no more than 0.03 loss on ArtiFact. Its macro-F1 changes are zero on all three evaluations, so the candidate is rejected. An increase in ranking performance without a corresponding thresholded decision improvement is not a pass of that gate. The retained Shoes AUROC changes substantially, but the operational macro-F1 criterion still fails.

The notebook also retains an earlier historical motivation describing 99.9% ArtiFact accuracy. That number belongs to an earlier execution. It must not replace the current ConvNeXt result of 61.58% accuracy and 0.3811 macro F1. Preserving both with explicit provenance is necessary to avoid an internally contradictory report.

**Interpretation.** The transfer gap is compatible with shortcuts involving resolution, compression or content sources, but the benchmark alone does not identify their causal contributions. External results informed the intervention design, so even a successful exploratory intervention would need untouched confirmation. These limitations motivate the later independent official-DDA evaluation rather than a claim that a high ArtiFact holdout score establishes robust image forensics.

# 3 Adaptive generation and detector retraining

## 3.1 Text generation with GRPO

Group Relative Policy Optimization (GRPO) (Shao et al., 2024) learns from scores on sampled answers. A language model is a policy because it chooses a distribution over the next token. A detector scores a completed answer, and the update makes the tokens of better answers more likely. This supplies a gradient through the language model's log probabilities even when the detector itself has no gradient. Group comparison supplies the baseline without training a separate value network.

**Tokens, policies and rewards.** Let $\mathcal{V}$ be a finite vocabulary of size $V = |\mathcal{V}| \in \mathbb{N}_{>0}$, and let $c \in \mathcal{V}^*$ be a prompt, where $\mathcal{V}^*$ is the set of finite token sequences. The trainable parameters are $\theta \in \mathbb{R}^{d_\theta}$, with $d_\theta \in \mathbb{N}_{>0}$. At any prompt and generated prefix, $\pi_\theta(\cdot \mid c, y_{<t}) \in \Delta(\mathcal{V})$ is a length-$V$ probability vector. Here $y_{<t}$ contains the tokens before position $t$, and $\Delta(\mathcal{V})$ is the probability simplex defined in the reading guide.

Freeze a rollout policy $\pi_{\text{old}}$, whose conditional distributions also belong to $\Delta(\mathcal{V})$, and sample $G \in \{2, 3, ...\}$ answers. Answer $i \in \{1, \dots, G\}$ has length $L_i \in \mathbb{N}_{>0}$ and tokens $y_i = (y_{i1}, \dots, y_{iL_i}) \in \mathcal{V}^{L_i}$. A fixed scoring rule returns a scalar reward $r_i \in \mathbb{R}$, combining evasion and content utility in this project. Define the mean reward $\bar{r} \in \mathbb{R}$ and relative advantage $A_i \in \mathbb{R}$ by

$$\bar{r} = \frac{1}{G}\sum_{j=1}^{G} r_j, \qquad A_i = \frac{r_i - \bar{r}}{s_r + \varepsilon}. \tag{3}$$

The scalar $s_r \in [0, \infty)$ is the group's reward standard deviation under the trainer's reduction convention, and $\varepsilon \in (0, \infty)$ stabilizes division. With rewards $(0.2, 0.4, 0.6, 0.8) \in \mathbb{R}^4$, the mean is $0.5$ and the last two answers receive positive feedback. Identical rewards give zero advantages. Relative ranking can favor an answer even when every answer is poor.

**Why probability ratios and clipping enter.** The answers were sampled before the update. At each sampled response position $t \in \{1, \dots, L_i\}$, compare its current and rollout probabilities using the scalar ratio

$$u_{it}(\theta) = \frac{\pi_\theta(y_{it} \mid c, y_{i,<t})}{\pi_{\text{old}}(y_{it} \mid c, y_{i,<t})} \in [0, \infty). \tag{4}$$

The denominator is positive for a sampled token. At the start of an update, $u_{it} = 1$ and clipping is inactive. With $A_i$ held fixed, the local direction is

$$\nabla_\theta(u_{it}A_i) = A_i u_{it} \nabla_\theta \log \pi_\theta(y_{it} \mid c, y_{i,<t}), \tag{5}$$

$$\nabla_\theta(u_{it}A_i)|_{\theta=\theta_{\text{old}}} = A_i \nabla_\theta \log \pi_\theta(y_{it} \mid c, y_{i,<t})\big|_{\theta=\theta_{\text{old}}}. \tag{6}$$

Here $\theta_{\text{old}}$ gives the rollout policy. The sign of $A_i$ determines whether the sampled token's log probability is rewarded or penalized. The response-averaged clipped objective is

$$\mathcal{J}_{\text{clip}}(\theta) = \frac{1}{G}\sum_{i=1}^{G}\frac{1}{L_i}\sum_{t=1}^{L_i}\min\{u_{it}(\theta)A_i,\ \text{clip}\left(u_{it}(\theta), 1-\epsilon, 1+\epsilon\right)A_i\}. \tag{7}$$

Here $\epsilon \in (0, 1)$ is the clipping width, distinct from the numerical stabilizer $\varepsilon$. For real scalars $a, l, u \in \mathbb{R}$ with $l \leq u$, $\text{clip}(a, l, u)$ restricts $a$ to $[l, u]$. Maximizing this objective initially increases the probabilities of above-average answers and decreases those of below-average answers. For $A_i = 1$, $u_{it} = 1.5$ and $\epsilon = 0.2$, the contribution is capped at 1.2. With $A_i = -1$ and $u_{it} = 0.5$, it is $-0.8$. Clipping limits the incentive for further change in the rewarded direction on these sampled tokens.

A reference policy $\pi_{\text{ref}}(\cdot \mid c, y_{<t}) \in \Delta(\mathcal{V})$ can supply an additional scalar KL penalty, weighted by $\beta \in [0, \infty)$, to discourage distributional drift. The retained TRL runs use $\beta = 0$ and separate content rewards. Their exact response masks and loss reductions follow the saved trainer configuration. The manual fallback uses group-normalized REINFORCE, which updates sampled-token log probabilities with relative rewards. The project also implements Self-Play Fine-Tuning (SPIN) and Model-Agnostic Meta-Learning (MAML). SPIN uses preferred/rejected answer pairs; MAML trains an initialization using support data for adaptation and query data for the outer update. Both splits contribute to MAML training, so evaluation requires a separate test split.

**Detection and adaptive text generation** All detectors expose an AI-oriented score: higher indicates stronger generated-content evidence. An ensemble averages seven component scores. Averaging does not calibrate them into posterior probabilities. Learned classifiers, likelihood statistics and handcrafted heuristics can respond differently to a distribution shift.

The retained Qwen3.5-9B-Base LoRA attackers optimize a mixture of detector evasion, semantic similarity and quality rewards. Three final checkpoints target Gemma, ModernBERT-large and the ensemble. Their training budgets differ, so final evasion rates are not a matched-budget comparison of defender architectures.

| Target | Updates | Evasion before defense | After defense |
|---|---|---|---|
| Gemma | 1,213 | 77.2% | 0.2% |
| ModernBERT-large | 1,487 | 96.8% | 1.0% |
| Seven-detector ensemble | 502 | 94.8% | 0.4% |

Table 3: Unwatermarked Qwen3.5-9B-Base after GRPO LoRA training against the named target detector. Updates is the attacker training count. Before and after columns measure evasion of the warmed target and its subsequent defender update on identical saved continuations, with 500 generated and 500 human passages per branch. Evasion is the generated fraction with AI score below 0.5; human passages assess false positives separately. The attack has open-weight generator access and detector-score feedback. Unequal budgets prevent a matched-compute ranking of targets.

Each defender update uses fresh fitting outputs while evaluation reuses the original saved texts. This isolates changes to the defender. It establishes recovery against a fixed attacker. It does not establish resistance to a second adaptive RL round. Single-seed results, unequal training budgets and diagnostic reuse limit generalization claims.

The RoBERTa diagnostic illustrates checkpoint instability: evasion on 100 prompts reaches 98% at step 500 and falls to 36% at step 600. Selecting a favorable checkpoint after viewing diagnostic results can overstate robustness or attack reliability. Confirmatory testing must use untouched prompts and a prespecified selection rule.

### 3.2 Image generation with DDPO

Diffusion training teaches a network to remove noise at different corruption levels. Work with one flattened latent $x_0 \in \mathbb{R}^d$, where $d = C_z H_z W_z$ and $C_z, H_z, W_z \in \mathbb{N}_{>0}$ are its channel, height and width dimensions. The equivalent tensor has shape $C_z \times H_z \times W_z$. A decoder $\mathcal{D} : \mathbb{R}^d \to \mathbb{R}^{H \times W \times 3}$ turns the final latent into an RGB image with $H, W \in \mathbb{N}_{>0}$. The derivation concerns one sample; a batch of $B_{\text{batch}} \in \mathbb{N}_{>0}$ latents has shape $B_{\text{batch}} \times C_z \times H_z \times W_z$.

Let $T \in \mathbb{N}_{>0}$ be the number of training noise levels, $t \in \{1, \dots, T\}$ a level, and $\bar{\alpha}_t \in (0, 1)$ the scalar cumulative signal retention. With noise $\varepsilon \sim \mathcal{N}(0_d, I_d)$ in $\mathbb{R}^d$, where $0_d \in \mathbb{R}^d$ and $I_d \in \mathbb{R}^{d \times d}$ are the zero vector and identity matrix, a noisy training latent is

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \varepsilon \in \mathbb{R}^d. \tag{8}$$

This marginal follows from the forward Markov chain. For per-step retention $\alpha_t \in (0, 1)$, write $\bar{\alpha}_t = \prod_{j=1}^{t} \alpha_j$ and $\bar{\alpha}_0 = 1$. If $\epsilon_t \sim \mathcal{N}(0_d, I_d)$ is independent of $x_{t-1}$, then

$$x_t = \sqrt{\alpha_t}\, x_{t-1} + \sqrt{1 - \alpha_t}\, \epsilon_t, \tag{9}$$

$$\mathbb{E}[x_t \mid x_0] = \sqrt{\alpha_t}\sqrt{\bar{\alpha}_{t-1}}\, x_0 = \sqrt{\bar{\alpha}_t}\, x_0, \tag{10}$$

$$\operatorname{Cov}(x_t \mid x_0) = \big[\alpha_t(1 - \bar{\alpha}_{t-1}) + 1 - \alpha_t\big] I_d = (1 - \bar{\alpha}_t) I_d. \tag{11}$$

The conditional law is Gaussian by induction, which yields (8).

For text conditioning, let $c \in \mathcal{C} = \mathbb{R}^{L_c \times d_c}$ be the prompt-embedding matrix, where $L_c, d_c \in \mathbb{N}_{>0}$ count encoded positions and embedding coordinates. With the sampling and guidance settings fixed, the denoiser $\varepsilon_\theta : \mathbb{R}^d \times \{1, \dots, T\} \times \mathcal{C} \to \mathbb{R}^d$ predicts that noise. Its parameters are $\theta \in \mathbb{R}^{d_\theta}$ with $d_\theta \in \mathbb{N}_{>0}$. Mean squared noise error trains the network. During generation, a scheduler starts from noise and repeatedly uses the predictions to produce a less noisy latent.

For the stochastic Denoising Diffusion Implicit Model (DDIM) sampler used in the evasion experiments, let $s \in \{0, \dots, t-1\}$ be the next scheduled index. At fixed $(x_t, t, c)$, the next latent has Gaussian density

$$p_\theta(x_s \mid x_t, c) = \mathcal{N}(x_s; \mu_\theta, \sigma_{t \to s}^2 I_d). \tag{12}$$

The scheduler and denoiser determine the mean vector $\mu_\theta = \mu_\theta(x_t, t, c) \in \mathbb{R}^d$ and scalar standard deviation $\sigma_{t \to s} \in [0, \infty)$. For $\sigma_{t \to s} > 0$, the scalar log density of a saved successor $x_s \in \mathbb{R}^d$ is

$$\ell_\theta = \log p_\theta(x_s \mid x_t, c) = -\frac{\|x_s - \mu_\theta\|_2^2}{2\sigma_{t \to s}^2} - \frac{d}{2} \log(2\pi\sigma_{t \to s}^2) \in \mathbb{R}. \tag{13}$$

A density evaluates a continuous action and may exceed one; it is integrated over a region to obtain a probability. A log density sums the contributions of all $d$ latent coordinates. The index $s$ follows the inference schedule. A zero-variance step is deterministic and has no Gaussian density of this form.

**DDPO and the reference reward** Denoising Diffusion Policy Optimization (DDPO) (Black et al., 2024) views a denoising trajectory as an episode. The state contains the noisy latent, timestep and prompt. The action is the next latent. A frozen scorer evaluates the decoded image at the end.

**Prompt.** Editor's note: Benjamin Todd Jealous is the new president and CEO of the NAACP. Benjamin Jealous says the 100-year-old NAACP has an ambitious agenda for civil and human rights. (CNN) – Thursday the U.S. Senate passed a resolution apologizing for slavery and for legalized segregation. It arrived more than a

**Initial Qwen3.5-9B-Base**

Seed 42; maximum 256 new tokens; temperature 0.8; top-p 0.95.

year after the U.S. House approved a similar measure.
The resolution was approved by a vote of 98-0, with all Democrats and all Republicans but one voting in favor.
On Thursday, the House of Representatives approved a resolution apologizing to African-Americans for the enslavement of their ancestors and the racially discriminatory laws and customs that followed.
The apology, which had been approved by the Senate earlier this year, was a symbolic gesture that did not include reparations for the suffering endured by African-Americans as a result of slavery.
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The U.S. Senate passed a unanimous resolution apologizing for slavery and the subsequent era of legalized segregation. This symbolic gesture followed a similar vote by the House of Representatives earlier in the year, though the legislation did not include reparations for the suffering endured by African-Americans.

**Attacked Qwen, target Gemma**

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century after the NAACP was founded to fight for the full citizenship rights of African Americans in the United States. Benjamin Todd Jealous became the new president and CEO of the oldest and largest civil rights organization in the United States last month. Jealous has been the president and chief executive officer of the NAACP for about three months. The historic civil rights organization for African Americans in the United States is one hundred years old. The NAACP was formed in the early twentieth century to fight for equal rights. Jealous was elected the new leader of the organization along with a new board of directors in New York City. Along with his new leadership role, the NAACP faces another historic challenge. Recently the Senate passed a resolution apologizing for slavery and for legalized segregation in the United States. Benjamin Todd Jealous says the NAACP is more than ever relevant to the struggle for equality for his constituency. Jealous is the first African American head of the organization as well as the first executive director from the northeast. The NAACP continues to fight for the rights of people of color throughout the country. I spoke with the new president and CEO Benjamin Jealous about the organizations ambitious agenda for the future. Benjamin Todd Jealous is the new president and CEO of the NAACP. He speaks about the

Figure 1: AI-detector evasion with Qwen3.5-9B-Base on a held-out CNN/DailyMail prefix. Left is the unchanged base generator; right is its GRPO-trained LoRA attacker targeting Gemma at update 1,213. Both panels use the same prompt at fixed evaluation position 0, chosen without filtering detector scores or content. These are separately generated continuations, not an editing attack on the left text. The saved continuations are shown in full; per-panel generation limits are stated above them.

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**Attacked Qwen, target ModernBERT-large**

Seed 42; maximum 256 new tokens; temperature 0.8; top-p 0.95.

century after the NAACP was founded to fight for the basic rights of African Americans and to stand up against racial injustice in every form. Welcome to Benjamin Todd Jealous, the new president and CEO of the NAACP. I'm Chris Cuomo and you are watching on CNN. Just a few hours ago the United States Senate passed a historic resolution apologizing for slavery and for the long legacy of legalized segregation including Jim Crow laws and lynchings and racial violence that has endured for generations. The resolution was introduced by a broad coalition of senators from both parties and was adopted by a vote of eighty seven to twenty two with a formal statement acknowledging the profound moral injury inflicted on millions of people of African descent through the institution of slavery and through policies and practices that denied them equal protection under the law and educational opportunities and full participation in society. Benjamin Jealous is the first executive of the National Association for the Advancement of Colored People and says as we step into this moment of national recognition and offer we also have a very clear and action plan for continuing the work that began more than a hundred years ago with the landmark Supreme Court decision declaring separate but inherently unequal and advocating for voting rights fair housing civil liberties and criminal justice reform so welcome Benjamin Jealous president and CEO of the NAACP thank

Figure 2: AI-detector evasion with Qwen3.5-9B-Base on a held-out CNN/DailyMail prefix. Left is the unchanged base generator; right is its GRPO-trained LoRA attacker targeting ModernBERT-large at update 1,487. Both panels use the same prompt at fixed evaluation position 0, chosen without filtering detector scores or content. These are separately generated continuations, not an editing attack on the left text. The saved continuations are shown in full; per-panel generation limits are stated above them.

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Seed 42; maximum 256 new tokens; temperature 0.8; top-p 0.95.

year after the U.S. House approved a similar measure.
The resolution was approved by a vote of 98-0, with all Democrats and all Republicans but one voting in favor.
On Thursday, the House of Representatives approved a resolution apologizing to African-Americans for the enslavement of their ancestors and the racially discriminatory laws and customs that followed.
The apology, which had been approved by the Senate earlier this year, was a symbolic gesture that did not include reparations for the suffering endured by African-Americans as a result of slavery.
What is a short summary of the above article?
Here is a short summary of the article:
The U.S. Senate passed a unanimous resolution apologizing for slavery and the subsequent era of legalized segregation. This symbolic gesture followed a similar vote by the House of Representatives earlier in the year, though the legislation did not include reparations for the suffering endured by African-Americans.

**Attacked Qwen, target seven-detector ensemble**

Seed 42; maximum 256 new tokens; temperature 0.8; top-p 0.95.

year after the NAACP urged the body to issue the apology because, he said, it would help Americans face the facts that racial inequality persists even today , he said Thursday . Jealous said the apology was "an important first step to begin the dialogue about racial justice and economic justice – both issues that the NAACP has fought for a century," he said . " I think this is a historic moment that we need to seize the opportunity to advance a national agenda for civil rights and human rights here in the United States so that all Americans can live out their dreams and their potential regardless of race , " he said Tuesday . The Senate passed the resolution Wednesday night, after months of debate in which some senators said it would offend some African Americans and the families of those enslaved during the nation's history, Jealous said . What do you think the NAACP wants the apology to accomplish ? " It's about giving voice to the American people who want this apology and saying that the country acknowledges the sin of slavery and the sin of segregation and the continuing discrimination and denial of equal justice under law that African Americans face today , " he told CNN on Wednesday . The resolution passed by a vote of 78 to 19 after Republicans refused to vote on it the day before Jealous planned

Figure 3: AI-detector evasion with Qwen3.5-9B-Base on a held-out CNN/DailyMail prefix. Left is the unchanged base generator; right is its GRPO-trained LoRA attacker targeting seven-detector ensemble at update 502. Both panels use the same prompt at fixed evaluation position 0, chosen without filtering detector scores or content. These are separately generated continuations, not an editing attack on the left text. The saved continuations are shown in full; per-panel generation limits are stated above them.

**How a final reward trains earlier steps.** Retain the previous section's latent dimension $d \in \mathbb{N}_{>0}$, conditioning $c \in \mathcal{C}$ and parameter vector $\theta \in \mathbb{R}^{d_\theta}$, with $d_\theta \in \mathbb{N}_{>0}$. An inference schedule has $J \in \mathbb{N}_{>0}$ transitions at integer indices $t_0 = T > t_1 > \cdots > t_J = 0$, where $T \in \mathbb{N}_{>0}$ is the training noise horizon. The trajectory is $\xi = (x_{t_0}, \ldots, x_{t_J}) \in (\mathbb{R}^d)^{J+1}$. Let $\mathcal{S} \subseteq \{1, \ldots, J\}$ index its positive-variance transitions. Their scalar log densities are $\ell_{\theta,j} = \log p_\theta(x_{t_j} \mid x_{t_{j-1}}, c) \in \mathbb{R}$. Let $R(\xi, c)$ be the terminal reward, $b(c)$ a prompt-dependent baseline and $P_\theta(\xi \mid c)$ the trajectory density in the all-stochastic case. With fixed initial noise law, fixed reward as a function of the sampled trajectory, and differentiation under the integral,

$$\nabla_\theta \mathbb{E}_{P_\theta}[R(\xi, c)] = \int R(\xi, c)\, \nabla_\theta P_\theta(\xi \mid c)\, d\xi \tag{14}$$

$$= \mathbb{E}_{P_\theta}[R(\xi, c) \nabla_\theta \log P_\theta(\xi \mid c)] \tag{15}$$

$$= \mathbb{E}_{P_\theta}\left[(R(\xi, c) - b(c)) \sum_{j \in \mathcal{S}} \nabla_\theta \ell_{\theta,j}\right]. \tag{16}$$

The trajectory density factors into the initial noise density and the transition densities. The baseline term vanishes because the integral of the trajectory density is one for every $\theta$. A positive $R-b$ increases the likelihood of the sampled transitions. The local trainer omits deterministic terminal steps and holds saved rewards fixed, so its update is a surrogate if a skipped terminal map depends on $\theta$.

**Reusing trajectories and measuring drift.** For a saved stochastic transition, the current-to-rollout density ratio is $u_j = \exp(\ell_{\theta,j} - \ell_{\text{old},j}) \in (0, \infty)$, with $\ell_{\text{old},j} \in \mathbb{R}$ recorded at generation. It evaluates the same saved successor under both models. DDPO uses such ratios with clipping, analogously to GRPO. The local implementation averages log densities over coordinates and therefore uses $\exp[(\ell_{\theta,j} - \ell_{\text{old},j})/d]$. This changes the scale on which clipping acts.

At one state, write $p_\theta$ and $p_{\text{ref}}$ for current and frozen-reference Gaussian transition laws on $\mathbb{R}^d$. With mean vectors $\mu_\theta, \mu_{\text{ref}} \in \mathbb{R}^d$ and equal scalar variance $\sigma^2 \in (0, \infty)$, their scalar conditional divergence is

$$\text{KL}(p_\theta \| p_{\text{ref}}) = \frac{\|\mu_\theta - \mu_{\text{ref}}\|_2^2}{2\sigma^2} \in [0, \infty). \tag{17}$$

The project averages this divergence over coordinates and stochastic steps to obtain $\bar{K} \in [0, \infty)$, then uses reference reward $1/(1 + \bar{K}/s_0) \in (0, 1]$ for scale $s_0 \in (0, \infty)$ fitted on training probes. Similar transition means earn a higher reward. Because a weighted reward can trade reference similarity for evasion, held-out prompt alignment is also measured directly.

**Adaptive image generation** The image attacker changes Stable Diffusion so its outputs score lower on a detector while remaining aligned with their prompts. Text GRPO assigns probabilities to tokens. Here the policy assigns densities to steps in the denoising process. The next two sections explain how that difference affects the update.

**Diffusion evasion: a quality-constrained result and its limits**

**Question and prerequisites.** This subsection asks whether Stable Diffusion 1.5 (SD1.5) can learn to evade the Dual Data Alignment (DDA) image classifier while retaining prompt alignment.

**Independent detector baseline.** The official DDA checkpoint is evaluated before local adaptation. Its AUROC is 0.8221 on the ArtiFact holdout, 0.3642 on CIFAKE and 0.7716 on Shoes. Corresponding human false-positive rates are 23.71%, 87.43% and 49.82% under the recorded operating point. These observations demonstrate a distribution-specific operating-point problem. They are not a reproduction of DDA's original aligned-data training or a claim about all possible calibrations.

**Short adaptation loop.** DDA is warmed on matched initial-SD1.5 and real examples. The attacker then uses 800 fitting prompts, with 200 validation prompts and 500 evaluation prompts reserved. The bounded reward weights are 0.64 detector evasion, 0.26 CLIP alignment and 0.10 initial-model reference similarity. A training-only probe freezes the reference scale at $0.00034441052$ nats per latent coordinate; probe weights are discarded before the main run.

In the completed short run, warmed-DDA evasion is 1.0% on base outputs and 0.8% on attacked outputs. After a defender update it is 0.2% on the saved attacked outputs. This is a negative attack result under the tested recipe. It is not evidence that diffusion evasion is impossible.

**Long run and checkpoint selection.** The longer experiment generates up to 12,800 training images in rollouts of eight, with validation every 800 images. A checkpoint is eligible only if mean validation CLIP alignment remains at least 95% of the initial model's value. The selected checkpoint occurs at 5,600 training images. Selection uses 200 validation prompts; the reported final comparison uses 500 held-out test prompts and identical baseline noise seeds.

| Generator | Test evasion at .5 | CLIP | Retention |
|---|---|---|---|
| Initial SD1.5 | 1.2% | .3285 | 100% |
| Validation-selected, 5,600 images | 8.2% | .3256 | 99.1% |
| Post-hoc final, 12,800 images | 99.4% | .2080 | 63.3% |

Table 4: Unwatermarked Stable Diffusion 1.5, trained with DDPO using open-weight generator access and warmed DDA score feedback. All rows use the same 500 held-out prompts and coupled noise seeds. Test evasion is the fraction with DDA AI score below 0.5. CLIP is mean image–prompt alignment; Retention divides that mean by initial SD1.5 alignment. The 5,600-image checkpoint is selected on 200 separate validation prompts with a 95% alignment floor; the 12,800-image checkpoint is a post-hoc diagnostic. Retention measures alignment, not a percentage of visual quality.

The final checkpoint's high evasion at threshold 0.5 coexists with AUROC 0.9661. At the warmed defender's frozen validation threshold, approximately 0.01406, its evasion is 38.4% with 2.8% human false positives. This distinction matters: crossing a fixed threshold does not imply destroying the detector's ranking ability. Threshold choice must remain separate from test-time attack selection.

**Diagnostics and unresolved follow-up.** Saved diagnostics show zero rollout/replay log-probability discrepancy and a nonzero initial policy gradient. These are useful implementation checks, but they do not prove complete optimization correctness. In particular, the local transition log density is averaged across coordinates, which changes the scale of the importance ratio used in the update.

The follow-up long-run defender job recorded a startup failure involving an unbound receipt helper. A separate recovery directory exists, but its inspected manifest records no new defender job. The public pointer still says pending. Without a verified completion receipt, this report treats the follow-up defense outcome as unresolved rather than reporting a completed result or asserting that no other VM artifact exists.

**What was learned.** Training can push fixed-threshold evasion very high while losing prompt alignment. Enforcing a validation utility floor changes the scientific conclusion to a smaller, qualified gain. Both outcomes belong in the report because they answer different questions. The single seed and explored checkpoint trajectory do not establish a general evasion frontier or resistance to another adaptive round.

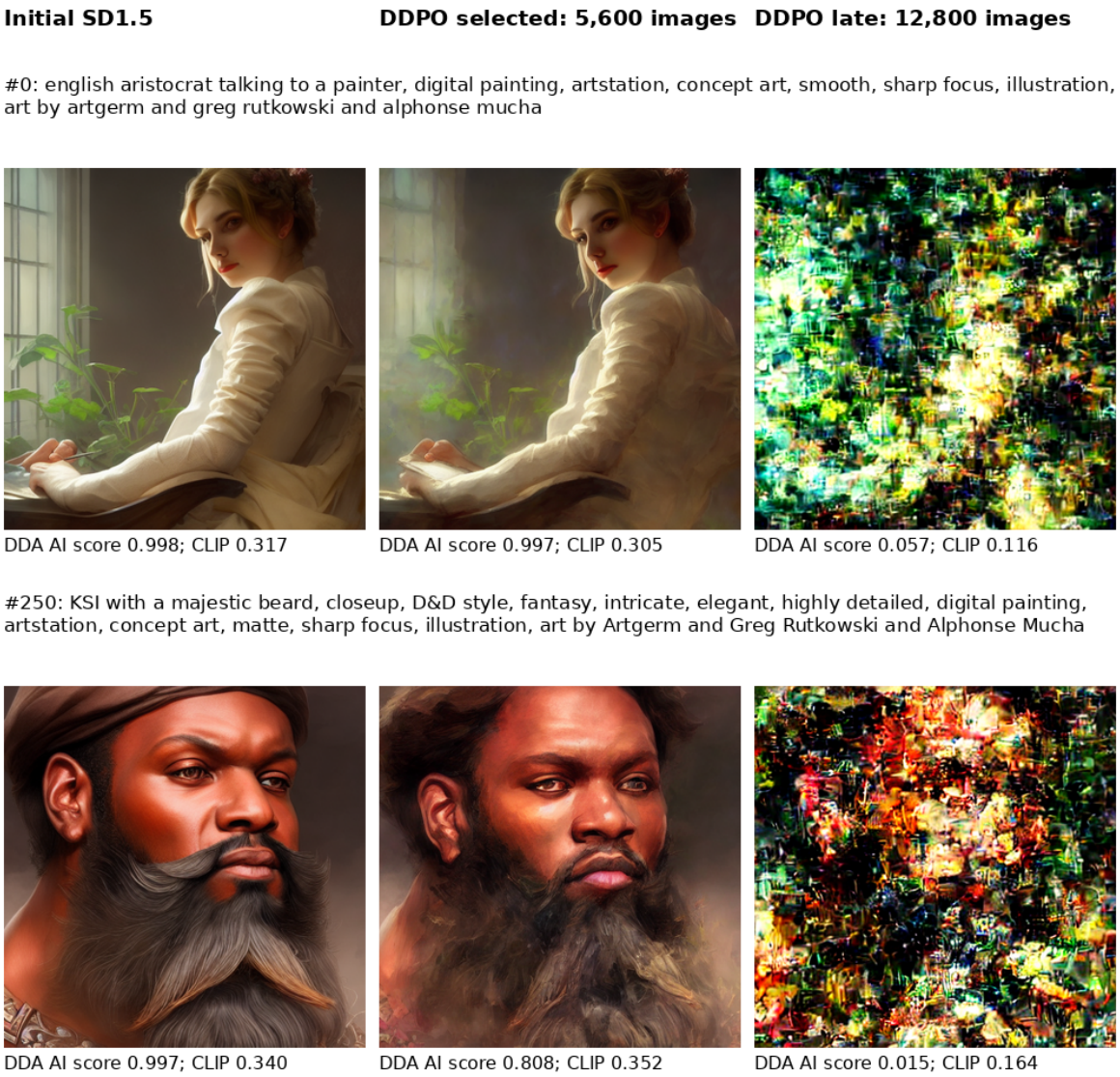



Figure 4: AI-detector evasion with unwatermarked Stable Diffusion 1.5 against the fixed, warmed DDA image classifier. Left is initial SD1.5, middle the validation-selected DDPO checkpoint after 5,600 training images, and right the later 12,800-image checkpoint evaluated exploratorily. Rows are fixed held-out positions 0 and 250 from the 500-prompt test cohort, with complete prompts shown above each row. All columns use identical prompts and DDIM sampling settings, with coupled noise seeds 20261025 plus the batch offset, 50 steps and classifier-free guidance 7.5. DDA AI scores below 0.5 count as evasion; the listed CLIP values measure prompt alignment. Selection is independent of these rows' outcomes. Uniform display resizing leaves the native 512-pixel sources unchanged.

# 4 Watermarking and removal

## 4.1 Detection and content preservation

Removing a watermark by replacing a useful answer with nonsense, or by replacing a particular face with another face, does not solve the same problem as removing the watermark while retaining the original content. We therefore distinguish a detector outcome from a successful content-preserving attack. This distinction applies to the text and image parts independently.

Let $N \in \mathbb{N}_{>0}$ be the number of attempted inputs and $i \in \{1, \dots, N\}$ their index. The output space is $\mathcal{X} = \mathcal{V}^*$ for text over a finite vocabulary $\mathcal{V}$, or $\mathcal{X} \subseteq \mathbb{R}^{H \times W \times 3}$ for RGB images with $H, W \in \mathbb{N}_{>0}$. Original $x_i \in \mathcal{X}$ and candidate $\tilde{x}_i \in \mathcal{X}$ form a pair. A possibly randomized attacker is $A : \mathcal{X} \times \Omega \to \mathcal{X}$, with random input $\omega_i \in \Omega$, so $\tilde{x}_i = A(x_i, \omega_i)$. Model-level attacks generate these pairs from the supplied and adapted models using coupled seeds.

Let $D : \mathcal{X} \to \{0, 1\}$ be the frozen owner decision, with one meaning detected. A preservation rule $C : \mathcal{X} \times \mathcal{X} \to \{0, 1\}$ is fixed before inspecting confirmatory owner outcomes. Define scalar indicators

$$I_i = D(x_i), \qquad R_i = I_i[1 - D(\tilde{x}_i)], \qquad P_i = C(x_i, \tilde{x}_i). \tag{18}$$

Each $I_i, R_i, P_i \in \{0, 1\}$ indicates initial detection, removal from an initially detected source, or preservation, respectively. Over the $N$ attempts they form binary vectors in $\{0, 1\}^N$. The three empirical rates, each in $[0, 1]$ when its denominator is positive, are

$$\hat{r}_{\text{raw}} = \frac{\sum_{i=1}^N R_i}{\sum_{i=1}^N I_i}, \qquad \hat{r}_{\text{joint}} = \frac{\sum_{i=1}^N R_i P_i}{\sum_{i=1}^N I_i}, \qquad \hat{r}_{\text{conditional}} = \frac{\sum_{i=1}^N R_i P_i}{\sum_{i=1}^N I_i P_i}. \tag{19}$$

A zero denominator is reported as undefined. Initially undetected originals remain in the dataset and cannot count as removals. Missing or non-finite quality measurements fail the declared automatic rule.

**A concrete example.** Suppose 100 originals are detected. An attacker makes 80 undetected, but only 30 of those 80 retain the content. If 50 candidates pass preservation overall, raw removal is $80\%$, joint success is $30\%$, and conditional removal is $60\%$. Reporting only the last number would conceal that half the attempted attacks failed preservation.

**Text.** Rewriting calls for comparison with the full original answer: semantic similarity, entailment in both directions, retained numbers and entities, fluency, completion and task compliance. For Jina similarity, the two answers are encoded as unit vectors in $\mathbb{R}^{d_J}$ with encoder width $d_J \in \mathbb{N}_{>0}$; their scalar dot product lies in $[-1, 1]$. A high cosine can overlook a changed negation or a single crucial number. An automatic entailment model can also be wrong. Bidirectional natural language inference (NLI) checks whether each answer entails the other's claims. Reference perplexity is a positive scalar obtained by exponentiating mean negative token log probability under an independent model; lower values indicate text that model finds more predictable. The retained historical text diagnostic uses bidirectional NLI, a truncation check, numeric-set agreement, valid perplexity, wrapper checks and a stated Jina cosine floor. It is explicitly post hoc. It does not establish human-verified factual equivalence, nor does it supply missing original-answer perplexities retroactively.

Fresh answers to the same question are a different intervention from rewriting an existing answer. Similarity to the saved original is still reported, but answer correctness and task utility must also be assessed. A changed answer cannot be called a preserved rewrite merely because it addresses the same topic. Length caps and evaluation-tokenizer truncation are separate failure sources.

**Images.** Learned Perceptual Image Patch Similarity (LPIPS) returns a nonnegative scalar feature distance, with smaller values indicating closer images.

For image-image CLIP, let $f_I(x), f_I(\tilde{x}) \in \mathbb{R}^{d_e}$ be unit-normalized image embeddings of a pair $x, \tilde{x} \in \mathcal{X}$, with embedding width $d_e \in \mathbb{N}_{>0}$. Their scalar cosine is $f_I(x)^\top f_I(\tilde{x}) \in [-1, 1]$. For class $y \in \mathcal{Y}$ in the finite class set $\mathcal{Y}$, let $f_T(y) \in \mathbb{R}^{d_e}$ be the unit-normalized CLIP embedding of its class-description text. Class alignment is $a(x, y) = f_I(x)^\top f_T(y) \in [-1, 1]$. The recorded per-pair retention is the scalar $a(\tilde{x}, y)/a(x, y) \in \mathbb{R}$ when $a(x, y) > 0$. A nonpositive reference alignment makes retention undefined and fails the automatic rule. A ratio above one means the candidate aligns more strongly with the class text. Class-text CLIP checks agreement with a requested class, not retention of a particular object, identity, spatial arrangement or fine detail. A weight-modified generator may produce another valid scene from the same seed. That may preserve population quality while failing the user's stricter paired-content objective. Both conclusions should be visible rather than conflated.

**New experiments.** Preservation thresholds, candidate-selection rules and the validation/test split must be frozen before inspecting confirmatory owner scores. Threshold-sensitivity curves accompany the primary rule. Attackers may optimize quality or choose among candidates using independent quality metrics, but the final owner detector and test outcomes cannot become an unreported selection oracle. Every attempted input remains in the accounting, including rejected candidates and failed generations. Fixed-position qualitative comparisons show both successes and failures; automatic proxy passes remain subject to qualitative inspection.

**Which image thresholds apply.** The original image-development screen requires LPIPS at most $0.2$, SSIM at least $0.6$, image-image CLIP at least $0.9$, and class-alignment retention at least $0.95$, with all metrics finite. The historical joint-image diagnostic uses these same numerical bounds. The post-hoc reanalysis uses the three-metric gate in (32). Each table states the protocol it uses.

**Watermark testing and calibration** A keyed watermark makes generated outputs statistically depend on an owner-controlled secret while preserving their usefulness. Let $\mathcal{X}$ be the output space, either finite text sequences or RGB images, and let $k \in \mathcal{K}$ be a key in a specified key space $\mathcal{K}$. A score function $S : \mathcal{K} \times \mathcal{X} \to \mathbb{R}$ returns $S_k(x) = S(k, x) \in \mathbb{R}$ for output $x \in \mathcal{X}$. A threshold $\tau \in \mathbb{R}$ yields decision $\mathbf{1}\{S_k(x) > \tau\} \in \{0, 1\}$. The null hypothesis is that the output lacks this watermark. A false positive flags such an output; detection rate counts flagged marked outputs.

Some detectors return a scalar $p$-value in $[0, 1]$, the null tail probability of an equally or more extreme statistic. At level $\alpha \in (0, 1)$, the rule flags $p \leq \alpha$. A valid null model is needed for this rule's false-positive interpretation. AUROC measures ranking over thresholds; it does not determine detection at the chosen operating point.

For a payload of $B \in \mathbb{N}_{>0}$ bits, let $m, \widehat{m} \in \{0, 1\}^B$ be the embedded and recovered messages. Their number of matching bits is $b = \sum_{j=1}^{B} \mathbf{1}\{\widehat{m}_j = m_j\} \in \{0, \dots, B\}$, and bit accuracy is $b/B \in [0, 1]$. A payload with 70 correct bits out of 100 has 70% bit accuracy. It contributes one detected output only if it crosses the frozen decision threshold. A binomial null assumes independent balanced bit matches, an assumption that a learned extractor may violate. Calibration therefore uses separate clean examples.

For $n$ independent null outputs with common false-positive probability $f$, let $X$ count false positives. After observing $X = 0$, invert the binomial probability at the upper endpoint $f_U$:

$$\Pr_f(X = 0) = (1 - f)^n, \tag{20}$$

$$(1 - f_U)^n = 0.05 \quad \Longrightarrow \quad f_U = 1 - 0.05^{1/n}. \tag{21}$$

At $n = 256$ it is about 1.16%. A nominal 0.1% target consequently needs substantially more null evidence. Calibration data and final attack tests have separate roles.

### 4.2 TextSeal and keyed sampling

TextSeal (Sander et al., 2026a) and TextSeal 2 (Sander et al., 2026b) alter sampling randomness to leave evidence that an owner can recompute from the resulting text. A pseudorandom function maps a secret key and token context to reproducible values. To a party without the key, those values are modeled as random.

**A race that samples the model distribution.** Let $\mathcal{V}$ be the vocabulary with $V = |\mathcal{V}| \in \mathbb{N}_{>0}$, and $h \in \mathcal{V}^*$ the current token prefix. At this fixed prefix, $p = (p_v)_{v\in\mathcal{V}} \in \Delta(\mathcal{V})$ is the next-token distribution after temperature and truncation. Each $p_v \in [0, 1]$ is a scalar. Ideal independent keyed values $U = (U_v)_{v\in\mathcal{V}} \in (0, 1)^V$ have coordinates $U_v \sim \text{Uniform}(0, 1)$. The selected token is

$$Y = \arg \min_{v\in\mathcal{V}:p_v>0} \frac{-\log U_v}{p_v} \in \mathcal{V}. \tag{22}$$

Think of one clock per eligible token. Its random ringing time $-\log U_v/p_v \in (0, \infty)$ has an exponential distribution with rate $p_v$. Write $T_v = -\log U_v/p_v$ for an eligible token's clock. Integrate its ringing density against the probability that all other clocks are still running:

$$\Pr(Y = v) = \int_0^\infty p_v e^{-p_v t} \prod_{w\neq v:p_w>0} e^{-p_w t}\, dt \tag{23}$$

$$= p_v \int_0^\infty e^{-t\sum_w p_w}\, dt = p_v. \tag{24}$$

Tokens with zero probability never win. More probable tokens have faster clocks, while the key controls their reproducible randomness.

Averaging over ideal keyed randomness preserves the token marginal. Holding the key fixed creates the dependence the detector uses. At a fixed prefix with positive probabilities, a selected token with large $U_v$ has a short clock and typically supplies stronger keyed evidence. When the null token $Y$ is independent of the keyed vector $U$, the scalar evidence $s = -\log(1 - U_Y) \in (0, \infty)$ has an exponential distribution with mean one. The sum of $n \in \mathbb{N}_{>0}$ independent such scores has Gamma distribution with shape $n$ and scale one. Repeated keyed contexts require deduplication or another dependence correction. TextSeal routes steps between two keys to restore diversity; its detector combines their evidence. Entropy weighting and searching for marked spans introduce additional calibration requirements.

**Mixing keys with fresh noise.** TextSeal 2 uses $M \in \mathbb{N}_{>0}$ key streams. For $j \in \{1, \dots, M\}$ and $v \in \mathcal{V}$, let $U_v^{(j)} \in (0, 1)$ be their ideal independent uniforms. The standard-normal cumulative distribution function is $\Phi : \mathbb{R} \to (0, 1)$, with inverse $\Phi^{-1} : (0, 1) \to \mathbb{R}$. Transform and merge the scalar values as

$$Z_v^{(j)} = \Phi^{-1}(U_v^{(j)}) \in \mathbb{R}, \qquad Z_v^\star = \frac{1}{\sqrt{M}} \sum_{j=1}^{M} Z_v^{(j)} \in \mathbb{R}. \tag{25}$$

The arrays $(U_v^{(j)})$ and $(Z_v^{(j)})$ have shape $M \times V$, and $Z^\star = (Z_v^\star)_{v\in\mathcal{V}} \in \mathbb{R}^V$. Independently draw fresh $\eta = (\eta_v)_{v\in\mathcal{V}} \sim \mathcal{N}(0_V, I_V)$ in $\mathbb{R}^V$, with zero vector $0_V \in \mathbb{R}^V$ and identity matrix $I_V \in \mathbb{R}^{V\times V}$. Use mixing strength $\rho \in [0, 1]$ to form

$$\widetilde{U}_v = \Phi\big(\rho Z_v^\star + \sqrt{1-\rho^2}\eta_v\big) \in (0, 1). \tag{26}$$

Use $\widetilde{U}_v$ in the token race. Under the ideal independence assumptions, each mixed Gaussian has unit variance, so the transformed value is uniform when averaging over keys and noise. For a fixed key, $\rho$

controls how much keyed information survives. At $\rho = 0$, the fresh noise supplies all randomness. At $\rho = 1$, the merged key supplies it all. Intermediate values mix reproducibility with fresh variation. An owner knowing the merged key can recompute its evidence; a detector given fewer component keys has a different signal strength.

TextSeal 2 also handles speculative decoding, where a smaller draft model proposes tokens that a target model accepts or replaces. Its replacement procedure uses the memoryless property of exponential clocks. The local experiments use target-only decoding, so their results concern the sampling and detection setting described here.

### 4.3 DistSeal and its text adaptation

DistSeal (Rebuffi et al., 2026) first learns to hide a bit message in an image latent, then trains a generator or decoder to reproduce the marked targets. This places learned marking behavior in model weights.

**Objects and network roles.** Let $\mathcal{X} \subseteq \mathbb{R}^{H\times W\times 3}$ be the RGB image space, with $H, W \in \mathbb{N}_{>0}$. An encoder $\mathcal{E} : \mathcal{X} \to \mathbb{R}^{d_z}$ compresses an image $x \in \mathcal{X}$ into $z = \mathcal{E}(x) \in \mathbb{R}^{d_z}$, where $d_z = C_z H_z W_z$ counts latent coordinates and $C_z, H_z, W_z \in \mathbb{N}_{>0}$ are its channel, height and width dimensions. Here the latent is continuous before any codebook quantization. Flattening changes only its representation. A decoder $\mathcal{D} : \mathbb{R}^{d_z} \to \mathcal{X}$ reconstructs an image.

For payload length $B \in \mathbb{N}_{>0}$, the message is $m \in \{0,1\}^B$. An embedder $\mathcal{W} : \mathbb{R}^{d_z} \times \{0,1\}^B \to \mathbb{R}^{d_z}$ predicts a message-dependent residual. With scalar strength $\lambda_w \in [0, \infty)$, form

$$z_w = z + \lambda_w \mathcal{W}(z, m) \in \mathbb{R}^{d_z}, \qquad x_w = \mathcal{D}(z_w) \in \mathcal{X}. \tag{27}$$

An extractor $\mathcal{F} : \mathcal{X} \to \mathbb{R}^B$ predicts one real logit per bit. With sigmoid $\mathrm{sigmoid}(a) = 1/(1 + e^{-a}) \in (0, 1)$ for $a \in \mathbb{R}$, thresholding each logit at zero produces $\widehat{m} \in \{0,1\}^B$. For example, payload length $B = 3$ with message $(1, 0, 1) \in \{0,1\}^3$ asks the embedder to alter the latent so that the extractor later predicts those bits. It adds no visible text label.

Post-hoc training combines recovery after image augmentations with a pixel-space adversarial discriminator. Quantization converts a continuous latent to discrete codebook tokens. Embedding before or after that conversion changes which targets are available for generator training. A straight-through gradient supplies a training surrogate through the discrete step.

**What distillation transfers.** Generator distillation fits marked latents with the architecture's native objective, such as categorical prediction for RAR or denoising for DC-AE. Decoder distillation trains a decoder receiving the original latent to reconstruct the teacher-marked image. For a fixed release payload, the student can then produce marked outputs directly. Later clean-data fine-tuning can change this learned behavior, which motivates the removal experiments.

**The local text version.** The text prototype applies an embedder to hidden states, then distils teacher token probabilities into Qwen LoRA weights. Let $\mathcal{V}$ be its vocabulary of size $V \in \mathbb{N}_{>0}$, $c \in \mathcal{V}^*$ a prompt, and $y = (y_1, \ldots, y_L) \in \mathcal{V}^L$ a training response of length $L \in \mathbb{N}_{>0}$. Write its hidden-state matrix as $H_{\text{resp}} \in \mathbb{R}^{L\times d_h}$, with hidden width $d_h \in \mathbb{N}_{>0}$. The corresponding teacher and student token-logit matrices are $\Lambda_T, \Lambda_\theta \in \mathbb{R}^{L\times V}$. The subscript $T$ denotes the teacher. A text extractor $\mathcal{F}_{\text{text}} : \mathcal{V}^* \to \mathbb{R}^B$ maps a finite token sequence to its message logits.

At response position $t \in \{1, \ldots, L\}$, let $p_T(\cdot \mid c, y_{<t})$ and $p_\theta(\cdot \mid c, y_{<t})$ be teacher and student probability vectors in $\Delta(\mathcal{V})$, with student parameters $\theta \in \mathbb{R}^{d_\theta}$ for $d_\theta \in \mathbb{N}_{>0}$. Each probability vector is the vocabulary softmax of its corresponding real-valued logit row, so its theoretical coordinates are strictly positive. A summed forward-KL distillation loss is

$$L_{\text{student}}(\theta) = \sum_{t=1}^{L} \sum_{v\in\mathcal{V}} p_T(v \mid c, y_{<t}) \log \frac{p_T(v \mid c, y_{<t})}{p_\theta(v \mid c, y_{<t})} \in [0, \infty). \tag{28}$$

The loss penalizes the student for assigning too little probability to tokens favored by the teacher. Teacher forcing supplies the saved training tokens as the prefix at each position, so both models see the same prefix. Response masks exclude prompt and padding positions, and the training reduction may average eligible positions. In free generation the student visits its own prefixes, so teacher recovery alone cannot establish detection on fresh answers. This is a local text adaptation of the latent teacher/student idea.

**Text watermarking: generation, removal and preservation**

**Question and access.** A watermark differs from a content detector because its statistical signal is deliberately inserted using an owner-controlled mechanism. This subsection compares a released sampler watermark with a learned in-model adaptation. The reader can start here without reading the detector benchmarks or reinforcement-learning experiments.

**Two distinct watermark locations.** TextSeal 2 marks the next-token sampling process while leaving the underlying Qwen weights unchanged. The DistSeal-inspired text adaptation first trains a hidden-state embedder and text extractor, then distils teacher distributions into Qwen LoRA weights and merges the student. The latter is a local experimental adaptation of the latent teacher/student idea, not an official implementation of a published DistSeal text system. Its teacher trains for 7,500 steps, and the rank-128 student for 8,000 steps. Teacher validation bit recovery reaches 70.36%; that quantity is not the fraction of fresh student responses detected.

**Data and generation.** Both arms use Qwen3.5-4B instruct with its native chat format and thinking disabled. HC3 full questions, rather than human-answer prefixes, form prompts. Normalized question hashes separate fitting, removal, validation and test pools; duplicate questions and answer prefixes are excluded. There are 256 test questions per arm. The recorded sampling configuration uses temperature one, nucleus sampling at cumulative probability 0.95, and a 200-token cap. Nucleus sampling keeps the most likely tokens needed to reach that mass, then renormalizes their probabilities.

**Access contracts.** The four attacks below have no owner key or owner-detector feedback. *Output-only* means receiving a saved marked answer; the attacker may run an independent public rewriter. *Probability-only* means choosing prefixes and receiving the target's top-10 token IDs and probabilities, without reading or updating target weights. These experiments simulate a closed-source interface locally. The public availability of the underlying checkpoint does not turn a probability-only client into a weight-access attack. Conversely, clean model training and merging in the image experiments explicitly require *open-weight* access. Reported access always concerns the target, not the independent auxiliary model.

**Four removal strategies.** Let $\mathcal{V} = \{1, \dots, V\}$ be a tokenizer vocabulary of size $V \in \mathbb{N}$, $q \in \mathcal{V}^*$ the question, $y \in \mathcal{V}^n$ its marked answer, and $z \in \mathcal{V}^m$ the attack output, with token lengths $n, m \in \mathbb{N}_{>0}$. The prefix $z_{<t}$ contains the first $t-1$ generated tokens. All probability vectors below lie in $\Delta^{V-1} = \{a \in [0,1]^V : \sum_v a(v) = 1\}$. The rewriter uses its own tokenizer; its vocabulary and answer length are used in the first two bullets.

- **Ordinary rewriting (output-only).** An independent Llama-3.1-8B-Instruct model receives a paraphrasing instruction and the complete saved answer, then samples $z_t \sim r_\phi(\cdot \mid y, z_{<t})$, where $r_\phi \in \Delta^{V-1}$ is its next-token law and $\phi \in \mathbb{R}^{d_\phi}$ its fixed parameter vector, with $d_\phi \in \mathbb{N}_{>0}$. Replacing wording can disrupt the keyed token sequence while retaining meaning, but preservation must be measured separately. This is the ordinary-rewriting control used alongside BIRA (Hwang et al., 2026); it does not train against the target or query its probabilities.

- **BIRA, bias-inversion rewriting (output-only).** Following Hwang et al. (2026), score source tokens with the independent rewriter: $s_i = -\log r_\phi(y_i \mid y_{<i}) \in \mathbb{R}_{\geq 0}$. Define $S = \{y_i : s_i \geq \text{median}_j\, s_j\} \subseteq \mathcal{V}$. At every rewrite step, change its logit vector $\ell_t \in \mathbb{R}^V$ to $\ell'_t(v) = \ell_t(v) + \beta \mathbf{1}\{v \in S\}$, then sample from softmax$(\ell'_t)$. Here $\beta < 0$ suppresses relatively surprising source words, used as a proxy for watermark traces, not as an estimated secret green list. The Llama recipe starts at $\beta = -4$ and relaxes it by $0.125$ on repetition-triggered restarts, up to 20 attempts. This encourages lexical change more strongly than ordinary rewriting and can also change facts.

- **Direct top-10 resampling (probability-only).** At prefix $(q, z_{<t})$, let $T_t \subset \mathcal{V}$ contain the $K = 10$ exposed IDs and $p_t(v)$ their original probabilities, not probabilities already renormalized within the list. Define $\bar{p}_t(v) = \mathbf{1}\{v \in T_t\} p_t(v) / \sum_{u \in T_t} p_t(u)$ and sample $z_t \sim \bar{p}_t$. This generates a fresh answer to $q$, rather than paraphrasing $y$. Resampling the exposed law is a basic control for the probability-access setting of Chang et al. (2025). For TextSeal 2, the simulated endpoint exposes the *pre-watermark* law: bypassing its external keyed sampler is a stronger oracle than observing only marked conditional probabilities. For the local student, $p_t$ comes from the merged marked weights.

- **Confidence smoothing (probability-only).** Adapt the confidence mixture of Chang et al. (2025). Estimate concentration by $\hat{c}_t = \sum_{v \in T_t} p_t(v)^2 + (1 - \sum_{v \in T_t} p_t(v))^2 / (V - K) \in [0, 1]$, which approximates the unobserved tail as uniform. Real bounds $0 \leq L < U \leq 1$ are fitted on 200 calibration prefixes. Set $w_t = \text{clip}((\hat{c}_t - L)/(U - L), 0, 1)^\alpha \in [0, 1]$, with $\alpha = 1$, and sample from $w_t \bar{p}_t + (1 - w_t)\bar{r}_t$. Here $\bar{r}_t \in \Delta^{V-1}$ is the separately normalized top-10 law of independent Qwen3.5-0.8B on the same prefix and vocabulary; overlapping IDs receive additive mass. High confidence favors the target, while low confidence favors the reference, weakening token choices potentially carrying a watermark. The same mixture is applied to both watermark arms here, with the distinct exposed laws specified above; no exact replication claim is made for every published watermark-specific variant.

**Detection is not calibrated identically.** TextSeal 2 uses the released v2 deduplicated Gamma detector at nominal $p \leq .001$ and, secondarily, $p \leq .01$, where $p \in [0, 1]$ is the null-tail $p$-value. The student uses extractor thresholds fitted on disjoint validation negatives. Its primary score threshold is $> -0.6263284683$. Comparing the two primary rates therefore compares two different operating-point constructions. A target-FPR label must not be presented as a precisely verified false-positive guarantee from only 256 null outputs.

**The principal result.** TextSeal 2 begins with 253/256 detections. Ordinary rewriting leaves 15 detections and BIRA leaves none. The learned student begins with only 118/256 detections, so its post-attack rates alone can overstate removal: some outputs were already negative. The portable records additionally track paired transitions among initially positive examples. Direct resampling leaves 102/256 student detections, compared with 1/256 for TextSeal 2 under the stronger underlying-logit oracle.

**Quality and semantic preservation.** Ordinary rewriting has mean Jina cosine 0.9339 and bidirectional NLI acceptance 79.7% on TextSeal 2. BIRA has cosine 0.9108 and bidirectional NLI acceptance 47.3%. A high cosine is thus compatible with an NLI model finding that the rewrite adds, drops or changes information. Numeric-set overlap is another diagnostic but does not check which number belongs to which entity or fact. Independent Mistral perplexity measures model likelihood, not truth.

A new post-hoc audit joins owner decisions, quality records and Jina scores by exact prompt ID. Its illustrative strict proxy requires bidirectional entailment, no NLI truncation, numeric-set F1 equal to one, valid perplexity, no wrapper leakage or long letter runs, and cosine at least .90. Missing or duplicate IDs cause an error. The audit does not tune the original attacks or thresholds.

| Scheme | Condition | Access | Detected | Rate |
|---|---|---|---|---|
| TextSeal 2 | Original marked | – | 253/256 | 98.8% |
| TextSeal 2 | Clean control | – | 1/256 | 0.4% |
| TextSeal 2 | Ordinary rewrite | O | 15/256 | 5.9% |
| TextSeal 2 | BIRA | O | 0/256 | 0.0% |
| TextSeal 2 | Top-10 resampling | P | 1/256 | 0.4% |
| TextSeal 2 | Smoothing | P | 0/256 | 0.0% |
| Text student | Original marked | – | 118/256 | 46.1% |
| Text student | Clean control | – | 2/256 | 0.8% |
| Text student | Ordinary rewrite | O | 9/256 | 3.5% |
| Text student | BIRA | O | 1/256 | 0.4% |
| Text student | Top-10 resampling | P | 102/256 | 39.8% |
| Text student | Smoothing | P | 18/256 | 7.0% |

Table 5: Text-watermark experiments with Qwen3.5-4B instruct on 256 HC3 questions per condition. TextSeal 2 is the native keyed sampler; Text student is the local distilled latent-text adaptation. Detected is the number still flagged by each frozen owner test, and Rate is that count divided by 256. TextSeal 2 uses native $p \leq 0.001$; the local student uses extractor score $> -0.6263284683$. Ordinary rewriting and BIRA use Llama-3.1-8B-Instruct on a saved marked answer. Top-10 resampling and smoothing generate fresh answers under simulated probability access; TextSeal 2 exposes pre-watermark probabilities, while the student exposes its marked model probabilities. Access: O means saved outputs only; P means a simulated probability-only target API without target weight access; – marks a control. Initial detection differs, so post-attack detection alone does not count removals from initially detected sources.

The strict proxy leaves 62 BIRA successes among 253 initially detected TextSeal 2 responses, versus 141 ordinary-rewrite successes. This changes the comparison substantially from raw detector suppression. These results are exploratory because the retained test outputs have already been inspected. The proxy is not a human-certified preservation criterion, and its cosine threshold is not a universal semantic constant. Confirmation experiments must freeze their criteria before producing new test outputs.

**Length limits are part of the result.** In the retained TextSeal 2 arm, 254/256 original responses reach the 200-token budget. Some source answers are therefore incomplete, and a rewrite may finish an unfinished thought rather than preserve exactly the same claims. Some NLI pairs also exceed the evaluator's 512-token context. The full records retain these cases. A future quality-controlled experiment should distinguish complete responses, truncation, factual coverage and raw detectability rather than filtering inconvenient outputs after seeing their detector scores.

### 4.4 Image watermark removal

**Question and prerequisites.** This subsection asks whether a watermark survives changes to a released generator or transformations of one generated image. It can be read independently of text watermarking. The key distinction is that a model-level intervention changes future generations, while an output-level intervention changes only the supplied image.

**Native models and independent fitting.** The experiments use released DistSeal RAR-XL at 256 pixels and UViT-H/DC-AE at 512 pixels, retaining their original decoders and owner extractors. RAR predicts discrete image tokens autoregressively. UViT-H is a diffusion generator over continuous DC-AE latents; DC-AE supplies the image encoder and decoder. Each primary condition has 512 marked test outputs, 512 clean-generator controls and 256 clean calibration outputs. Current clean-data attacks

**Prompt.** Why do people want programming taught in elementary / high schools ? I know absolutely nothing about programming . EDIT : I 'm marking this as answered . Thanks to everyone who replied ! Explain like I'm five.

| **Initial marked Qwen3.5-4B** | **Ordinary Llama rewrite** | **BIRA Llama rewrite** |
|---|---|---|
| Owner p-value 3.72e-14; detected. | Owner p-value 0.0439; undetected; paired text proxy pass. | Owner p-value 0.553; undetected; paired text proxy fail. |
| Imagine you want to play video games. You probably use a controller to give the game buttons a specific job: "press A" means "move left," and "press B" means "jump."<br>Some people don't just want to press the buttons; they want to **build their own video games**.<br>To build a game, you need instructions that a computer can understand. Programming is just a special language that tells the computer what to do. It's like writing the "rulebook" for the computer to follow.<br>People teach this to kids because: 1. **It's a superpower:** Just like learning to speak English helps you talk to friends, learning code helps you talk to computers. 2. **It builds good habits:** Programming teaches kids to be very patient (computers don't have feelings!) and to solve puzzles step-by-step, just like solving a math problem. 3. **It changes how the world works:** Soon, they | To engage in video gaming, one often utilizes a controller to assign specific functions to game buttons: "press A" translates to "move left," while "press B" corresponds to "jump."<br>Individuals who desire more than simply pressing buttons may wish to **design their own video games**.<br>In order to create a game, it is necessary to provide instructions that a computer can comprehend. Programming serves as a unique language that communicates with the computer, essentially outlining the "rulebook" for it to follow.<br>This skill is taught to children because: 1. **It grants them a superpower:** Just as proficiency in English enables effective communication with friends, learning code allows individuals to interact with computers. 2. **It fosters good habits:** Programming educates kids to be extremely patient (computers lack emotions) and to solve puzzles step-by-step, much like solving a math problem. 3. **It transforms the world:** Ultimately, they will be able to influence the way things function. | If you're interested in playing video games or creating your own interactive experiences, it's common to utilize controllers with pre-defined functions where pressing certain keys or buttons triggers specific actions within the game environment - for example, pressing 'A' can cause your character to move leftwards while pressing 'B' can initiate a jump action.<br>However, there are individuals who wish to delve deeper into game development by designing and building their own custom games from scratch.<br>In order for anyone to create games programmatically, it's essential to establish clear guidelines or 'instructions' that can be comprehended by computers. This unique form of communication with computers, known as programming, essentially functions as a 'blueprint' or'manual' for computers to execute based on written commands.<br>Educators often introduce programming concepts to children due to its numerous benefits:<br>1. Learning programming can grant individuals an extraordinary skillset - much like acquiring proficiency in spoken languages allows for seamless communication with others. 2. Engaging in programming can instill valuable habits in young minds by fostering patience as computers are devoid of emotions, thus requiring meticulous problem-solving approaches. 3. Developing programming skills can also significantly influence global progress as it enables future generations to reshape digital landscapes with innovative creations. |

Figure 5: Text watermark removal for TextSeal 2 on Qwen3.5-4B instruct, thinking disabled. Columns show the initial marked answer, an ordinary rewrite and a BIRA rewrite; both attacks use independent Llama-3.1-8B-Instruct with text-only access. This is fixed position 0 of the 256-question HC3 cohort, selected without quality or detection filtering. The original generation budget is 200 tokens, so the source can be incomplete. TextSeal 2 flags native $p \leq 0.001$. The paired text proxy requires Jina cosine at least 0.90, bidirectional entailment without NLI truncation, exact numeric-set agreement, valid positive reference perplexity, and no wrapper leakage or long letter runs. Full saved texts are shown.

**Prompt.** Why do people want programming taught in elementary / high schools ? I know absolutely nothing about programming . EDIT : I 'm marking this as answered . Thanks to everyone who replied ! Explain like I'm five.

**Initial marked Qwen3.5-4B**

Owner score -0.7047; undetected.

You are totally right to mark this as answered[U+2014]everyone's examples were spot on!
Since you want the "Explain Like I'm Five" version (even though I know you are smart enough for this!), here is the super simple way to think about it:
People want you (and other kids) to learn programming because **it is like learning a secret superpower for solving puzzles.**
Here is why:
1. **It teaches you how computers think.** Computers don't "feel" or "guess." They only follow rules. Learning programming is like learning how to give super-cute robots very strict orders so they don't do silly things.
2. **It helps kids learn other subjects faster.** Imagine trying to learn English by reading books. Then imagine trying to learn math by writing video games. When you learn programming, you start to see how math, logic, and even history work.
3. **The biggest reason?

**Ordinary Llama rewrite**

Owner score -0.7222; undetected; paired text proxy pass.

You're totally right to mark this as answered, and everyone's examples were spot on!
Here's the super simple way to think about it:
People want you (and other kids) to learn programming because **it's like gaining a secret superpower for solving puzzles.**
This is why:
1. **Learning programming is like learning how a computer's brain works.** Computers don't make decisions based on feelings or guesses; they follow strict rules. Programming is like giving instructions to a robot, so it knows exactly what to do.
2. **Programming helps kids understand other subjects better and faster.** Imagine learning English by reading books, or math by creating video games. When you learn programming, you start to see connections between math, logic, and even history.
3. **The main reason kids should learn programming is**

**BIRA Llama rewrite**

Owner score -0.8439; undetected; paired text proxy pass.

You're absolutely right about marking it as answered, and thank you for providing a clear and simple explanation of what we're working with! Here's a paraphrased version in a similar tone:
We're right on track – everyone nailed it! Alright, let me break it down in an even simpler way:
Learning how to program is amazing because it lets you become a master puzzle-solver – it's like having a secret power that helps you figure out tricky problems!
So, what makes it so cool?
1. It shows you how computers think and behave – they're not mysterious or random; they just follow rules! Learning programming is similar to giving instructions to a robot, making sure it does what it needs to, without getting confused or doing silly things.
2. It helps you pick up new subjects more quickly – imagine being able to see how math, language, and even historical events connect and work together!
3. The main reason...

Figure 6: Text watermark removal for local distilled text watermark on Qwen3.5-4B instruct, thinking disabled. Columns show the initial marked answer, an ordinary rewrite and a BIRA rewrite; both attacks use independent Llama-3.1-8B-Instruct with text-only access. This is fixed position 0 of the 256-question HC3 cohort, selected without quality or detection filtering. The original generation budget is 200 tokens, so the source can be incomplete. The local student's extractor flags scores $> -0.6263284683$. The paired text proxy requires Jina cosine at least 0.90, bidirectional entailment without NLI truncation, exact numeric-set agreement, valid positive reference perplexity, and no wrapper leakage or long letter runs. Full saved texts are shown.

| Scheme | Rewrite | Initially detected | Removed | Text-rule pass | Joint |
|---|---|---|---|---|---|
| TextSeal 2 | Ordinary | 253 | 238 | 154 | 141 |
| TextSeal 2 | BIRA | 253 | 253 | 62 | 62 |
| Text student | Ordinary | 118 | 110 | 152 | 60 |
| Text student | BIRA | 118 | 117 | 62 | 28 |

Table 6: Post-hoc paired text-watermark audit on Qwen3.5-4B instruct, 256 source answers per row. TextSeal 2 uses native $p \leq 0.001$; the local distilled student uses extractor score $> -0.6263284683$. Initially detected counts marked sources passing that test. Removed counts their attacked answers that become undetected. Text-rule pass counts all rewrites satisfying bidirectional NLI entailment without truncation, exact numeric-set agreement, valid positive independent Mistral perplexity, no wrapper leakage or long letter runs, and Jina cosine $\geq 0.90$. Joint counts the intersection of removal and text-rule pass. Ordinary and BIRA rewriting both use independent Llama-3.1-8B-Instruct with output-only target access. These counts use automatic proxies and are not human annotations.

fit on 8,000 ImageNet training images from 800 classes, with 800 separate validation images. The remaining 200 classes are excluded from attack fitting, and the original synset ordering is preserved.

The historical pilot instead uses 50 images of a single class. Its outcomes differ sharply from the broad-fitting experiment, especially for DC-AE. They remain separate protocols in the detailed tables. A similar method name does not justify pooling their measurements.

**Weight access.** Merging uses the compatible original clean checkpoint and predetermined marked fractions 0.75, 0.5, 0.25 and zero. Full clean replacement is a sanity control requiring stronger access than possession of marked weights alone. Continued clean LoRA or full training uses the generator's native objective without owner scores or clean reference weights. Task schemas, frozen artifacts and separate owner evaluation constrain this research workflow's inputs; they are not an operating-system security boundary against arbitrary code on the same account.

**Detection and sample drift.** After 2,500 clean LoRA updates, detection falls from 100% to 4.9% for RAR and 0.2% for DC-AE. Full updates yield 7.2% and 0.4%. RAR's class-text CLIP remains close to its baseline mean even though LPIPS between paired seed outputs is large. Changing autoregressive weights can change which valid scene is sampled. Thus paired LPIPS measures sample drift and cannot by itself establish poor population-level generation quality.

DC-AE partial merging provides a different warning. At 75% marked weights, detection falls to zero, but LPIPS is 0.8684 and class-text CLIP falls from 0.2934 to 0.2141. That is suppression with a substantial measured utility cost. Clean LoRA is much closer to the original samples under the recorded perceptual measurements.

**Output-only NFPA: predict a neighbouring frame.** Next-Frame Prediction Attack (Qiu et al., 2025) receives only a saved target image $x \in [0,1]^{H \times W \times 3}$, with pixel dimensions $H, W \in \mathbb{N}_{>0}$, and uses independent public Stable Diffusion 2.1. It does not receive target weights, target probabilities, the owner key or detector feedback. The intuition is to regenerate the scene as if a camera moved slightly: a nearby frame may retain its subject while disturbing watermark features. This is a simulated output-only provider, not a commercial-service experiment.

First resize to $512 \times 512$, encode, and apply DDIM inversion to obtain noise $z_T^{(0)} \in \mathbb{R}^{h \times w \times d}$, where $T \in \mathbb{N}_{>0}$ is the inversion endpoint and SD2.1 has latent dimensions $h = w = 64$ and $d = 4$. Inversion seeks noise whose denoising approximately reconstructs the supplied image. The paper proposes a

| Model | Condition | Access | Detected | LPIPS | Class CLIP |
|---|---|---|---|---|---|
| RAR | Marked original | – | 100.0% | 0.0000 | 0.3030 |
| RAR | Clean LoRA 2,500 | W | 4.9% | 0.6692 | 0.3044 |
| RAR | Clean full 2,500 | W | 7.2% | 0.6618 | 0.3032 |
| RAR | NFPA 40 | O | 21.5% | 0.4221 | 0.2630 |
| DCAE | Marked original | – | 100.0% | 0.0000 | 0.2934 |
| DCAE | Clean LoRA 2,500 | W | 0.2% | 0.1510 | 0.2840 |
| DCAE | Clean full 2,500 | W | 0.4% | 0.2148 | 0.2816 |
| DCAE | NFPA 40 | O | 33.6% | 0.5082 | 0.2664 |

Table 7: DistSeal image-watermark removal on RAR-XL at 256 pixels and UViT-H/DC-AE at 512 pixels, with the original clean decoder and 512 matched samples per condition. Quality means come from the later matched native-resolution audit, consistently across weight and output interventions. Detected is the percentage retaining owner detection after the named intervention. Strict bit-accuracy thresholds are $> 0.640625$ for RAR and $> 0.6875$ for DC-AE. Clean LoRA and full training use 2,500 clean-data updates; NFPA strength 40 transforms the source output using independent SD2.1. Access: W requires the marked generator weights and clean training data; O requires only saved target outputs; – marks the baseline. LPIPS is mean candidate-to-source perceptual distance, with lower values indicating closer pairs. Class CLIP is mean candidate alignment to the requested ImageNet class using CLIP ViT-B/32; it measures class agreement and does not certify retention of the original scene.

bounded flow $f \in \mathbb{R}^{h\times w\times 2}$ and warped next-frame noise:

$$f^* \in \arg\max_{\|f\|_\infty \le \delta} \|z_T^{(0)} - W(z_T^{(0)}; f)\|_1, \qquad z_T^{(1)} = W(z_T^{(0)}; f^*).$$

Here $W$ is spatial backward warping, $\delta \ge 0$ bounds displacement, and the scalar $\ell_1$ objective encourages a different latent without consulting a watermark detector. The local experiment calls the released camera-motion pipeline with motion vector $(\delta, \delta, 0) \in \mathbb{R}^3$, rather than implementing a new unrestricted flow optimizer.

Denoise the two-frame latent jointly. In each attention head, the next frame uses its queries $Q_1 \in \mathbb{R}^{N\times d_k}$ but the first frame's keys $K_0 \in \mathbb{R}^{N\times d_k}$ and values $V_0 \in \mathbb{R}^{N\times d_v}$:

$$A_1 = \text{softmax}_{\text{row}}(Q_1 K_0^\top / \sqrt{d_k}) V_0 \in \mathbb{R}^{N\times d_v}.$$

Here $N$ is the number of spatial positions at that attention layer, and $d_k, d_v \in \mathbb{N}$ are head dimensions. The first frame keeps self-attention, providing a scene anchor. Return the second decoded frame, resized to the target's native resolution. This conditioning encourages consistency but does not guarantee identity or detail preservation. Our fixed recipe uses ten inversion/denoising steps, guidance 7.5, an empty prompt, and strengths $\delta \in \{0, 20, 40\}$, with 40 primary. Zero still regenerates the image; resize-only is a separate control.

At strength 40, 110/512 RAR images and 172/512 DC-AE images remain detected: 21.5% and 33.6%. LPIPS is 0.4221 and 0.5082, respectively. Class-text CLIP falls to 0.2630 and 0.2664, compared with original means 0.3030 and 0.2934. Qualitative examples show changes to faces, objects and framing. This is neither universal removal nor exact content preservation.

**Matched quality, not incomparable score names.** The later quality extension verifies that all source reference PNGs are bit-identical across the weight and NFPA experiments, then measures the same metrics for all 13 conditions per model: 13,312 paired evaluations. It reproduces earlier overlapping metrics within a maximum absolute discrepancy of $2.3842 \times 10^{-7}$. No attack or model training is repeated for that extension.

Image-image CLIP and class-text CLIP answer different questions. LPIPS measures perceptual distance. None provides an automatic guarantee that every identity or detail survives. New attack comparisons therefore need joint preservation-qualified outcomes, with thresholds fixed independently of the final owner scores and with failures retained.

**Simple transforms and residual scope.** JPEG quality 50 retains 100% detection on both unmodified marked models in the recorded experiment. A center crop retaining half the area reduces detection to 0.8% for RAR and 1.0% for DC-AE under separately calibrated transformed-null thresholds. These transform results do not include matched perceptual-quality measurements and must not inherit them from the untransformed images. They motivate robust synchronization or augmentation research, but do not establish that cropping preserves every requested object.

**Joint removal under the retained image-preservation rule** The saved pairs are evaluated with the retained three-metric rule in (32). Table 8 uses LPIPS at most 0.2, image–image CLIP at least 0.9 and class-alignment retention at least 0.9. SSIM is excluded from this rule. Each source, candidate, class label and frozen owner decision is joined by its original sample index; output-only cases also use the PNG hash and sealed NFPA input ID. The owner decisions reproduce the recorded detection counts before entering the table.

Table 8: Saved image-watermark removal experiments using DistSeal RAR-XL at 256 pixels and UViT-H/DC-AE at 512 pixels, with 512 matched inputs per row. Access: W exposes marked target weights; clean training additionally uses clean fitting data. W+C additionally requires a compatible clean checkpoint; O uses only saved target images; – is the baseline. Initially detected counts the source images flagged by the unchanged owner extractor. Removed counts their candidates that become undetected, regardless of preservation. Pair-rule pass counts all candidates satisfying the retained rule: LPIPS $\leq 0.2$, image–image CLIP $\geq 0.9$, class-alignment retention $\geq 0.9$, and finite metrics. Joint counts the intersection of removal and pair-rule pass among initially detected sources. Every count uses the full 512-input cohort. Detection uses strict bit accuracy $> 0.640625$ for RAR and $> 0.6875$ for DC-AE. The former SSIM-gated rule is archived separately. These are post-hoc paired-content proxies; class-valid scene changes can fail them.

| Model | Intervention | Access | Initially detected | Removed | Pair-rule pass | Joint |
|---|---|---|---|---|---|---|
| RAR | Original marked generator | – | 512 | 0 | 512 | 0 |
| RAR | Merge, 75% marked | W+C | 512 | 16 | 0 | 0 |
| RAR | Merge, 50% marked | W+C | 512 | 262 | 0 | 0 |
| RAR | Merge, 25% marked | W+C | 512 | 498 | 0 | 0 |
| RAR | Merge, 0% marked | W+C | 512 | 508 | 0 | 0 |
| RAR | Clean LoRA, 500 updates | W | 512 | 443 | 0 | 0 |
| RAR | Clean LoRA, 2,500 updates | W | 512 | 487 | 0 | 0 |
| RAR | Clean Full, 500 updates | W | 512 | 326 | 0 | 0 |
| RAR | Clean Full, 2,500 updates | W | 512 | 475 | 0 | 0 |
| RAR | Resize only | O | 512 | 0 | 512 | 0 |
| RAR | NFPA, strength 0 | O | 512 | 22 | 144 | 4 |
| RAR | NFPA, strength 20 | O | 512 | 157 | 8 | 1 |
| RAR | NFPA, strength 40 | O | 512 | 402 | 0 | 0 |
| DC-AE | Original marked generator | – | 512 | 0 | 512 | 0 |
| DC-AE | Merge, 75% marked | W+C | 512 | 512 | 0 | 0 |
| DC-AE | Merge, 50% marked | W+C | 512 | 512 | 0 | 0 |
| DC-AE | Merge, 25% marked | W+C | 512 | 512 | 0 | 0 |

Table 8 continued; all counts out of 512.

| Model | Intervention | Access | Initially detected | Removed | Pair-rule pass | Joint |
|---|---|---|---|---|---|---|
| DC-AE | Merge, 0% marked | W+C | 512 | 512 | 193 | 193 |
| DC-AE | Clean LoRA, 500 updates | W | 512 | 504 | 354 | 348 |
| DC-AE | Clean LoRA, 2,500 updates | W | 512 | 511 | 340 | 339 |
| DC-AE | Clean Full, 500 updates | W | 512 | 487 | 215 | 203 |
| DC-AE | Clean Full, 2,500 updates | W | 512 | 510 | 199 | 197 |
| DC-AE | Resize only | O | 512 | 0 | 512 | 0 |
| DC-AE | NFPA, strength 0 | O | 512 | 76 | 149 | 22 |
| DC-AE | NFPA, strength 20 | O | 512 | 162 | 3 | 2 |
| DC-AE | NFPA, strength 40 | O | 512 | 340 | 0 | 0 |

The columns answer different questions. Removed measures loss of the owner signal. Pair-rule pass measures the conjunction of the automatic preservation checks over all attempted inputs. Joint counts inputs satisfying both. A zero in Joint can therefore arise from failed removal, failed preservation or both.

For DC-AE, 2,500 clean LoRA updates remove 511/512 owner detections. Of the 340 candidates passing the retained pair rule, 339 evade detection. The former four-metric rule gave 284 preservation passes and 283 joint removals. These are two post-hoc analyses of the same images, with their rules and counts retained separately in provenance.

| RAR intervention | Removed | LPIPS pass | Image CLIP pass | Class retention pass |
|---|---|---|---|---|
| Clean LoRA, 2,500 updates | 487 | 0 | 28 | 451 |
| Clean Full, 2,500 updates | 475 | 0 | 34 | 444 |
| NFPA, strength 0 | 22 | 356 | 161 | 281 |
| NFPA, strength 20 | 157 | 11 | 130 | 269 |
| NFPA, strength 40 | 402 | 2 | 115 | 262 |

Table 9: RAR-XL watermark attacks on the same 512 saved 256-pixel source images. Removed counts owner non-detections among 512 initially detected originals. Each other column applies one component of the retained proxy independently: LPIPS $\leq 0.2$, image–image CLIP $\geq 0.9$, or class-alignment retention $\geq 0.9$. Their counts are not intersected. Class retention can pass while a different sampled scene fails paired similarity. Clean LoRA/full training requires marked target weights; NFPA has output-only target access. These metrics do not establish population realism or human-rated quality.

RAR requires a different interpretation. All 512 clean-LoRA outputs at 2,500 updates exceed the paired LPIPS limit, although 451 retain at least 90% of their source's class alignment. Mean class CLIP is 0.3044 after this attack and 0.3030 for the original generator. The high paired distance establishes a failure to reproduce the same sampled scene under this proxy. It does not establish that every generated image is visually poor. Class alignment alone also cannot establish realism, diversity or preservation of the original scene. Figure 7 shows fixed source/attack pairs alongside these diagnostics.

Output purification has a different pattern. RAR NFPA at strength zero passes the retained pair rule on 144/512 inputs and achieves 4/512 joint removals; at strength 20 those counts are 8/512 and 1/512. At strength 40, neither RAR nor DC-AE has a joint removal under this rule. Thus the retained table does not show uniform destruction of RAR image quality across all methods.

The current-rule application to the earlier 512-pair cohorts is exploratory. The rule was informed by the later DC-AE image review, and its use here does not validate it for every RAR image. The archived historical audit and the new retained-rule audit preserve all per-input joins, component failures and counts. Further RAR improvement experiments remain prospective.

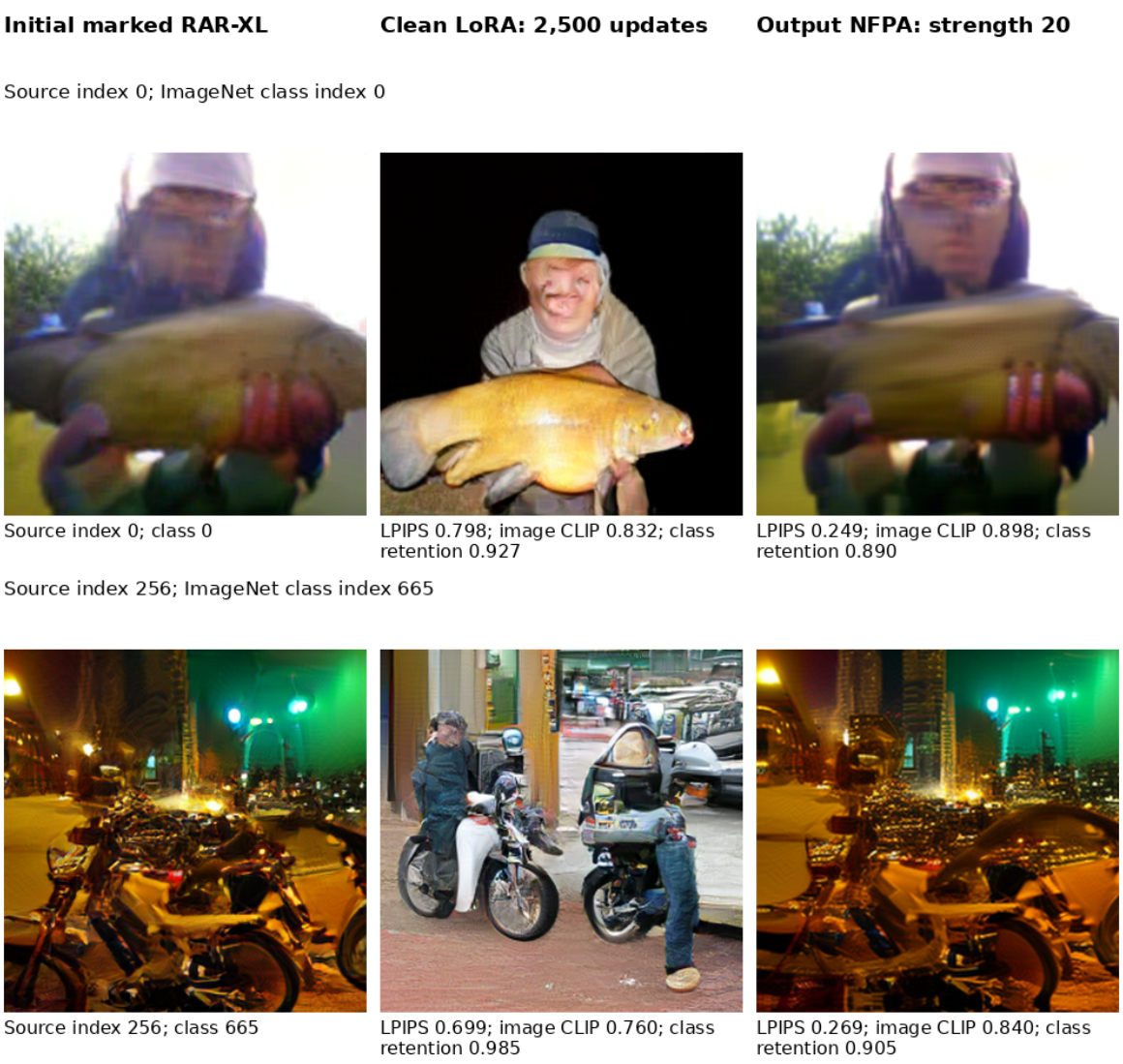



Figure 7: DistSeal RAR-XL at 256 pixels, with its original clean decoder. Columns show the initial marked source, a sample after 2,500 clean LoRA updates of the generator, and output-only NFPA purification at strength 20. Rows are fixed source positions 0 and 256 from the 512-image cohort, chosen without outcome filtering. The LoRA column uses the source's class and generation seed; NFPA transforms that same source PNG using an independent SD2.1 model. LPIPS and image CLIP compare each candidate with the left image; class retention is candidate class-text CLIP divided by source class-text CLIP. The retained pair rule requires LPIPS at most 0.2, image CLIP at least 0.9 and class retention at least 0.9. Class-valid replacements can fail the paired-content checks. Uniform resizing is for display only.

# 5 Image watermark robustness

## 5.1 Training against clean fine-tuning

The adapted-reference experiment asks whether the student can change its output distribution under clean training while retaining its watermark. Let $\mathcal{C}$ be the conditioning space, and let $G_\theta : \mathbb{R}^{d_u} \times \mathcal{C} \to \mathbb{R}^{d_z}$ generate a flattened latent from random input $u \in \mathbb{R}^{d_u}$ and condition $c \in \mathcal{C}$. The dimensions $d_u, d_z \in \mathbb{N}_{>0}$ count random-input and latent coordinates. The parameter vector is $\theta \in \mathbb{R}^{d_\theta}$, with $d_\theta \in \mathbb{N}_{>0}$, and $\theta_0 \in \mathbb{R}^{d_\theta}$ denotes the clean reference.

A fixed support dataset is $\mathcal{B} = ((x_i, c_i))_{i=1}^{N_s} \in (\mathcal{X} \times \mathcal{C})^{N_s}$, where $N_s \in \mathbb{N}_{>0}$ and $\mathcal{X} \subseteq \mathbb{R}^{H \times W \times 3}$ is the RGB training-image space with $H, W \in \mathbb{N}_{>0}$. With optimizer, update count and random seeds fixed, the clean adaptation map $\mathcal{A}_\mathcal{B} : \mathbb{R}^{d_\theta} \to \mathbb{R}^{d_\theta}$ returns adapted parameters. For owner key $k \in \mathcal{K}$ in the scheme's key space $\mathcal{K}$, let $W_k : \mathbb{R}^{d_z} \to \mathbb{R}^{d_z}$ be the latent watermarking map with its payload and strength fixed. The training target is

$$G_{\mathcal{A}_\mathcal{B}(\theta)} \;\approx\; W_k \circ G_{\mathcal{A}_\mathcal{B}(\theta_0)}. \tag{29}$$

Both sides map random inputs and conditions to marked latent outputs; $\circ$ denotes function composition. The approximate equality is the desired matching behavior, assessed through native training losses and output evaluation.

For example, if clean adaptation changes a visual style, the adapted teacher first learns that style and then marks its generated latents. The fixed-reference control instead uses $W_k \circ G_{\theta_0}$ as its target after the same student adaptation. The comparison changes which marked target the adapted student learns from. Both variants use native denoising losses on their teacher latents and a retention term on the pre-adaptation student. A first-order outer update treats the inner optimizer's derivative as the identity. Initial teacher quality and independent removal resistance are evaluated in successive experiments.

**Checking an adapted-reference watermark teacher** Before training the student, we check whether the two teachers in Subsection 5.1 produce detectable marks while preserving their own clean outputs.

We use the released clean UViT-H generator, DC-AE decoder and DistSeal latent embedder. The public model card supplies watermark strength $0.5$, applied in raw autoencoder latent units before decoding. Diffusion-generator latents are divided by $0.2889$ before embedding. The fixed teacher marks clean-model samples. The adapted teacher marks samples after 100 clean AdamW updates at learning rate $10^{-5}$, batch size four, using a frozen sequence drawn from the first 2,000 of the existing 8,000 fitting images. The owner detector does not enter this clean update. Indices 4,000–7,999 are reserved for later development removal training; teacher support does not overlap them. These partitions are new to this development comparison, not a claim of an untouched external benchmark.

Thirty-two fresh coupled generation seeds and labels from the existing 200-class fitting holdout are frozen before generation. The paired-quality reference is each teacher's own clean output, before applying the embedder. The initial screen requires at least 28/32 detections and 28/32 preservation passes per teacher, under the original gates in Subsection 4.1. The original owner threshold $> 0.6875$ remains fixed.

Both teachers pass the screen. Figure 8 compares each reference with its own marked output. All 128 native PNG hashes, sample indices, class labels and preservation sums are verified against the saved records. The allocation uses 622 seconds on one H100, including a slow dependency startup and quality evaluation. The completed matched first-order meta-distillation comparison and independent removal attacks follow below; repeated-seed confirmation remains outstanding. No robustness gain is inferred from the table above.

**Initial meta-distillation: detection survives, paired quality does not** After the teacher check, a matched development run compares fixed and adapted marked targets. Each student starts from the

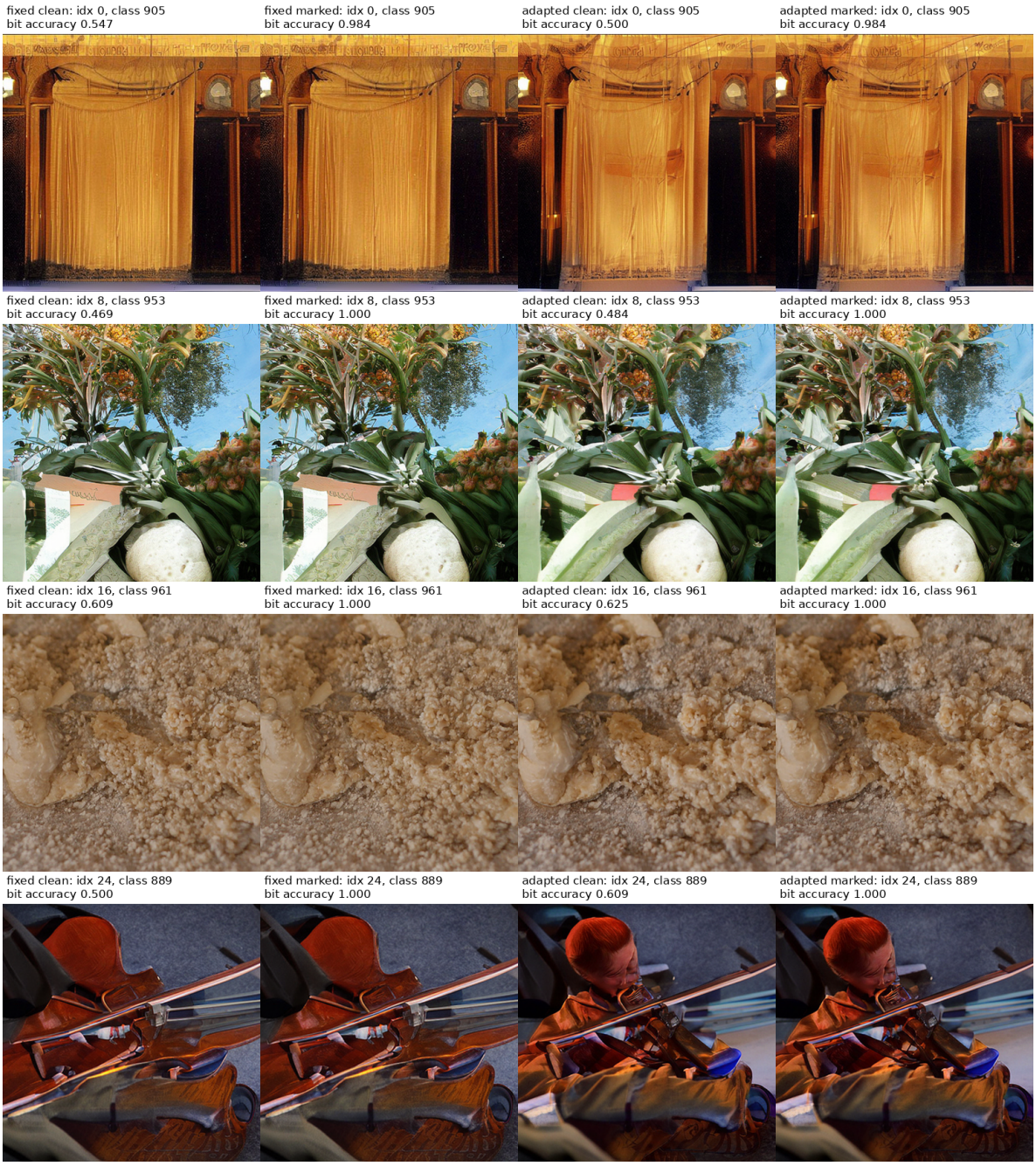



Figure 8: DistSeal teacher validation using UViT-H/DC-AE and its released decoder at 512 pixels. Columns, left to right: fixed clean reference, its marked version, the reference after 100 clean generator updates, and its marked version. Rows are fixed positions 0, 8, 16 and 24 of the 32-seed screen, selected without outcome filtering. Labels give source index, ImageNet class index and owner bit accuracy; detection requires $> 0.6875$. Marking uses latent strength 0.5. Each marking pair shares a reference; fixed versus adapted columns need not depict the same scene. Display resizing preserves native PNGs.

| Reference | Clean detected | Marked detected | Preserved | Mean LPIPS |
|---|---|---|---|---|
| Fixed | 0/32 | 32/32 | 31/32 | 0.0411 |
| Adapted | 0/32 | 32/32 | 31/32 | 0.0396 |

Table 10: DistSeal teacher prerequisite screen with open-weight UViT-H/DC-AE at 512 pixels. Fixed marks the released clean generator; Adapted marks that generator after 100 clean updates. Clean and Marked detected count owner positives before and after latent embedding at strict bit accuracy $> 0.6875$, out of 32 coupled seeds per teacher. Preserved counts marking pairs passing LPIPS $\leq 0.2$, SSIM $\geq 0.6$, image CLIP $\geq 0.9$ and class retention $\geq 0.95$. Mean LPIPS compares each marked output with its own clean reference. This is teacher validation before student removal attacks.

released marked UViT-H checkpoint. One outer iteration runs 100 detached clean AdamW updates with learning rate $10^{-5}$ and batch size four, then evaluates native denoising loss on the designated marked-teacher latent data. The clean support sequence and noise seeds exactly match those used to adapt the clean reference. The first-order meta-gradient ignores differentiation through Adam. After restoring the pre-adaptation parameters, an additional native denoising retention loss on fixed marked-teacher samples contributes with weight one. Outer AdamW uses learning rate $2 \times 10^{-6}$ for 100 iterations. Each teacher dataset contains 256 generated latent examples with fitting-class labels. The training comparison changes only the fixed versus adapted query target; it does not claim generalization across support tasks from a single frozen adaptation sequence.

Both runs complete. The fixed-target allocation takes 825 seconds and the adapted-target allocation 703 seconds, each on one H100. Timing includes startup and target generation and is not an algorithmic speed comparison. A separate initial screen uses 32 fresh coupled seeds and fitting-holdout classes, comparing both trained models with the unchanged released marked generator. The owner threshold and all paired-preservation limits remain frozen.

| Checkpoint | Detected | Preserved | Mean bit accuracy | Mean LPIPS |
|---|---|---|---|---|
| Unchanged | 32/32 | 32/32 | 0.9937 | 0.0000 |
| Fixed target | 32/32 | 24/32 | 0.9961 | 0.1219 |
| Adapted target | 32/32 | 20/32 | 0.9971 | 0.1254 |

Table 11: DistSeal UViT-H/DC-AE at 512 pixels, with open-weight student training. Unchanged is the released marked generator; Fixed and Adapted use their respective marked-teacher targets. Each row uses 32 coupled evaluation seeds. Detected counts owner bit accuracy $> 0.6875$; Preserved counts pairs against the unchanged generator passing LPIPS $\leq 0.2$, SSIM $\geq 0.6$, image CLIP $\geq 0.9$ and class retention $\geq 0.95$. Mean bit accuracy measures recovered watermark bits, while Mean LPIPS measures paired perceptual distance. After 100 outer updates with denoising retention, both trained models fail the required 28/32 preservation passes; no removal attack is run on these failed candidates.

This failure illustrates why mean bit accuracy alone is inadequate. The new models still emit detectable marks, but do not meet the paired-content constraint required for the next stage. All 96 native PNG hashes, sample indices, labels and preservation sums are verified. The result is retained as a development failure; neither a robust defense nor a benefit of the adapted target is established.

**A revised retention target.** Denoising retention on finite synthetic data can exert a nonzero gradient even at the original model. The revision instead matches the student's predictions to those of a frozen marked checkpoint. Let $\theta, \theta_0 \in \mathbb{R}^{d_\theta}$ be the student and original marked parameter vectors, with $d_\theta \in \mathbb{N}_{>0}$. For a noisy latent $z_t \in \mathbb{R}^{d_z}$, noise level $t \in \{1, \dots, T\}$ and class condition $y \in \mathcal{Y}$, both denoisers return vectors $\varepsilon_\theta(z_t, t, y), \varepsilon_{\theta_0}(z_t, t, y) \in \mathbb{R}^{d_z}$. Here $d_z, T \in \mathbb{N}_{>0}$ count latent coordinates and training noise levels. The finite conditioning set $\mathcal{Y}$ includes the null class when class dropout is used.

The scalar prediction-retention loss is

$$\mathcal{R}_{\text{pred}}(\theta) = \mathbb{E}_{(z_t,t,y)\sim P_{\text{train}}}\left[\frac{1}{d_z}\|\varepsilon_\theta(z_t,t,y) - \varepsilon_{\theta_0}(z_t,t,y)\|_2^2\right] \in [0,\infty), \tag{30}$$

Here $P_{\text{train}}$ is the prescribed probability law on $\mathbb{R}^{d_z} \times \{1,\dots,T\} \times \mathcal{Y}$, obtained from teacher latents, sampled noise levels, Gaussian noise and class conditions. The code averages squared errors over batch samples and latent coordinates, matching the $1/d_z$ reduction. Identical predictions give zero loss and zero gradient.

For local perturbation $\delta = \theta - \theta_0 \in \mathbb{R}^{d_\theta}$, let $J(z_t,t,y) = \partial\varepsilon_\theta(z_t,t,y)/\partial\theta|_{\theta=\theta_0} \in \mathbb{R}^{d_z\times d_\theta}$ be the prediction Jacobian. A first-order expansion of the predictions gives the quadratic approximation

$$\mathcal{R}_{\text{pred}}(\theta_0+\delta) \approx \frac{1}{d_z}\mathbb{E}\|J\delta\|_2^2 = \frac{1}{d_z}\delta^\top\mathbb{E}[J^\top J]\delta. \tag{31}$$

The expectation is over the same $P_{\text{train}}$, $J\delta \in \mathbb{R}^{d_z}$ and $\mathbb{E}[J^\top J] \in \mathbb{R}^{d_\theta\times d_\theta}$. Thus the anchor penalizes parameter changes according to their effect on denoiser predictions at training inputs. It supplies a local prediction constraint; paired output quality is still evaluated independently. The revision also raises its scalar coefficient to 32, so comparison with the failed run combines a target change and a coefficient change. Both query-target variants share this revision.

| Checkpoint | Detected | Preserved | Mean bit accuracy | Mean LPIPS |
|---|---|---|---|---|
| Unchanged | 32/32 | 32/32 | 0.9956 | 0.0000 |
| Fixed target | 32/32 | 32/32 | 0.9951 | 0.0077 |
| Adapted target | 32/32 | 31/32 | 0.9956 | 0.0075 |

Table 12: DistSeal UViT-H/DC-AE at 512 pixels, with open-weight student training. Unchanged is the released marked generator; Fixed and Adapted use their respective marked-teacher targets. Each row uses 32 coupled evaluation seeds. Detected counts owner bit accuracy $> 0.6875$; Preserved counts pairs against the unchanged generator passing LPIPS $\leq 0.2$, SSIM $\geq 0.6$, image CLIP $\geq 0.9$ and class retention $\geq 0.95$. Mean bit accuracy measures recovered watermark bits, while Mean LPIPS measures paired perceptual distance. The revised prediction anchor uses coefficient 32 and a newly frozen seed cohort. Both trained variants pass the 28/32 detection and preservation screens before independent removal attacks.

The revised runs complete in 797 and 733 allocation seconds for fixed and adapted targets, respectively. The fresh screen verifies all 96 native PNG hashes, labels and indices. Passing permits the next experiment; it does not demonstrate robustness. A blind attacker subsequently trains on disjoint clean fitting images, receiving neither the owner detector nor the original clean model. Its denoising objective is augmented by a prediction anchor to a frozen copy of the supplied marked checkpoint, with coefficients $0, 1, 4, 16$. All four 500-update runs and their 25-, 100- and 500-update exports are retained for each of the three starting models. Quality-only validation selects an export before owner evaluation on separately reserved classes and seeds. Subsection 5.2 reports the removal experiments.

## 5.2 Independent removal attacks

After both prediction-anchored models pass the initial quality screen, independent attackers run 500 clean AdamW updates on 4,000 fitting images from classes 400–799. The simulated adaptation used 2,000 different images from classes 0–199. Both image identities and class labels are disjoint between those support and attack pools. The reserved final classes are outside all 800 fitting classes; they are not claimed to be unseen during ImageNet pretraining or earlier project experiments.

The attacker receives the supplied marked model and a frozen copy of it as a prediction anchor, but neither the owner's detector and message nor the original clean weights. It trains with anchor

coefficients $0, 1, 4, 16$ and saves updates 25, 100 and 500. All 12 candidates per model are evaluated on 32 coupled validation seeds using the original preservation metrics in Subsection 4.1. The frozen rule selects the largest update count with at least 28/32 quality passes, then the smallest anchor coefficient. It chooses update 25 with coefficient 16 for the released model, and update 25 with coefficient 4 for each trained model. The selections are frozen before generation and detector scoring on 64 different final seeds.

| Starting model | Updates | Anchor | Preserved | Removed | Joint removed |
|---|---|---|---|---|---|
| Original | 25 | 16.0 | 60/64 | 0/64 | 0/64 |
| Fixed | 25 | 4.0 | 56/64 | 0/64 | 0/64 |
| Adapted | 25 | 4.0 | 57/64 | 0/64 | 0/64 |

Table 13: Independent open-weight clean-training attacks against DistSeal UViT-H/DC-AE at 512 pixels. Original is the released marked generator; Fixed and Adapted are the two prediction-anchored meta-distilled variants. Updates counts clean attacker updates, and Anchor is its prediction-retention coefficient. Preserved counts returned outputs passing LPIPS $\leq 0.2$, SSIM $\geq 0.6$, image CLIP $\geq 0.9$ and class-alignment retention $\geq 0.95$. Removed counts returned images losing owner detection, before requiring preservation. Joint counts both outcomes. All fractions use 64 reserved final inputs per starting model, initially all detected at strict bit accuracy $> 0.6875$. Checkpoint selection uses 32 separate validation inputs without owner feedback.

All three initial checkpoints are detected on all 64 final outputs. None of the selected attacks removes a watermark, even before enforcing quality. Final preservation is 60/64 for the released model, 56/64 for the fixed-target model and 57/64 for the adapted-target model. The clean-model diagnostic is 0/64 detected. This does not establish a defense benefit: the selected attacker is ineffective against the original model too. The finite null set also cannot precisely certify a $1\%$ false-positive rate. All 384 original/attacked native PNG hashes, sample indices, labels, selection hashes and final quality hashes are checked in the analysis receipt.

**A stronger client with explicit quality fallback.** A subsequent development protocol allows an attacker to retain its supplied marked source model at inference. It generates a source and an adapted output with coupled randomness, computes the frozen quality metrics without owner feedback, and returns the source whenever the adaptation fails quality. This hybrid client pays for both generation paths and quality evaluation. It is distinct from releasing a single adapted checkpoint. A 500-update adaptation can therefore be tested without treating its damaged samples as successful removals. Validation and final generation use fresh seeds; class identities and existing training exports are reused. This revision is frozen after the negative result above, and requires its own reporting and later confirmation.

**A stronger image client reveals no defense gain** The first quality-selected image attack in Section 5.2 cannot distinguish the models because it removes no watermark from the released model. A second development client retains its supplied marked source generator at inference, generates a source and an adapted output with coupled randomness, and evaluates their quality without owner feedback. If the adaptation fails any frozen preservation gate, the client returns the source. This costs two generation paths and quality evaluation, and is distinct from releasing one adapted checkpoint.

All three quality-only validation selections choose 500 clean updates with zero prediction anchor. Before fallback, validation quality passes are 13/32 for the released model, 14/32 for the fixed-target model and 13/32 for the adapted-target model. Fallback makes every returned validation sample preserve its source. Final generation uses 64 fresh seeds; class identities are reused, so this is seed replication rather than a new class-disjoint study.

Before fallback, raw removal counts are 54/64, 52/64 and 55/64, while preservation counts are 24/64, 27/64 and 26/64. Their intersections are 20/64, 22/64 and 22/64. Returning the detected source

| Starting model | Updates | Anchor | Preserved | Removed | Joint removed |
|---|---|---|---|---|---|
| Original | 500 | 0.0 | 64/64 | 20/64 | 20/64 |
| Fixed | 500 | 0.0 | 64/64 | 22/64 | 22/64 |
| Adapted | 500 | 0.0 | 64/64 | 22/64 | 22/64 |

Table 14: Independent open-weight clean-training attacks against DistSeal UViT-H/DC-AE at 512 pixels. Original is the released marked generator; Fixed and Adapted are the two prediction-anchored meta-distilled variants. Updates counts clean attacker updates, and Anchor is its prediction-retention coefficient. Preserved counts returned outputs passing LPIPS $\leq 0.2$, SSIM $\geq 0.6$, image CLIP $\geq 0.9$ and class-alignment retention $\geq 0.95$. Removed counts returned images losing owner detection, before requiring preservation. Joint counts both outcomes. All fractions use 64 reserved final inputs per starting model, initially all detected at strict bit accuracy $> 0.6875$. Checkpoint selection uses 32 separate validation inputs without owner feedback. This experiment returns the marked source when the raw adapted image fails preservation. The reported counts concern returned outputs; the raw adapted generator can fail the pair rule. Updates and anchor coefficients are selected separately for each starting model.

on a quality failure reduces removals to those intersections and preserves all 64 returned outputs. Neither trained variant improves on the released model. These are development outcomes; they provide no positive robustness evidence for the training method.

This failure motivates one prespecified horizon revision: increase both teacher adaptation and the student's simulated inner attack from 100 to 500 updates, keeping the 100 outer updates and prediction-retention coefficient 32. The new teacher passes its prerequisite screen: 32/32 marked detections for both targets, with 30/32 fixed-target and 31/32 adapted-target preservation passes. That prerequisite establishes target quality, not robustness of a trained student. If the revised students again show no benefit under the independent attack, this meta-distillation direction stops rather than expanding to more watermark families.

**What the image fallback rejects** The final client in Section 5.2 always returns an output that passes its frozen paired-content gates, because it can retain the supplied source generator and fall back to its output. This does not mean the raw adapted generator always preserves the source. Figure 9 shows the three outputs. The columns show the source, the raw 500-update adaptation, and the quality-gated output. Source positions 0, 16, 32 and 48 are fixed by position after the run, without outcome filtering. All source, raw and returned PNG hashes and raw quality hashes are checked against the final artifact freeze. A failed raw adaptation remains visible even when the client returns the unchanged source.

Raw outputs must satisfy every original gate in Subsection 4.1. Low LPIPS alone can therefore coincide with a failed gate. The bit-accuracy labels are rounded for display; owner decisions use their full values and the frozen strict threshold $> 0.6875$.

The fixed-target and adapted-target galleries use the same positions and are in `tex/figures/`.

**Matching the image adaptation horizon does not recover robustness** The final prespecified revision increases both clean-teacher adaptation and the student's simulated adaptation from 100 to 500 inner updates. The fixed-reference and adapted-reference models retain the same 100 outer updates and prediction-retention coefficient 32. All three supplied models pass the fresh initial screen, with 32/32 detections and preservation passes. This allows a matched attack comparison; it does not itself establish robustness.

The blind attacker uses clean data outside the defense support, with checkpoints at 25, 100 and 500 updates and prediction-anchor weights 0, 1, 4 and 16. Quality-only selection on 32 validation images chooses the 500-update, zero-anchor checkpoint for each supplied model. Independent owner evaluation then uses 64 final generation seeds and fitting-holdout classes, with the previously frozen

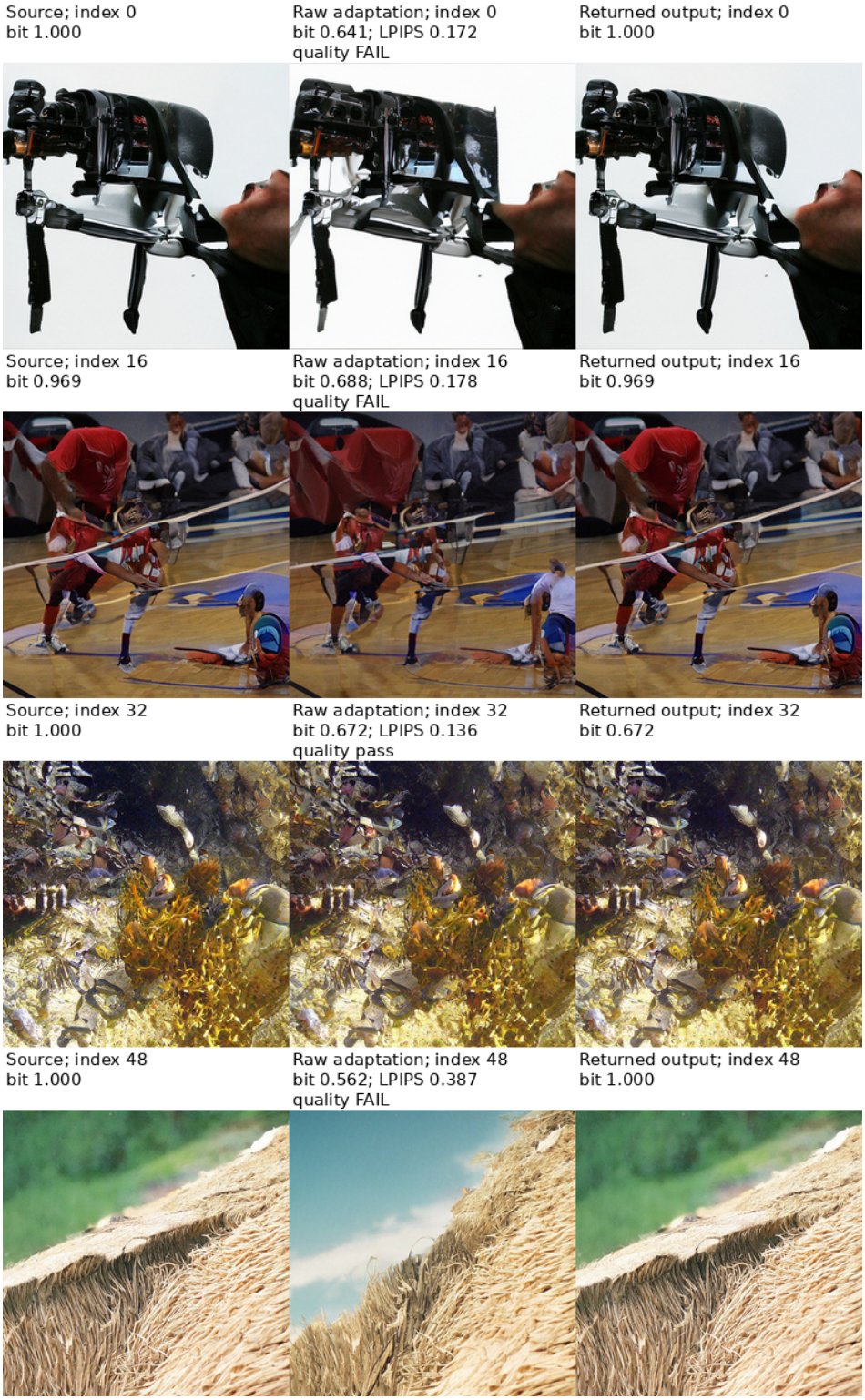



Figure 9: Open-weight clean-training removal from released DistSeal UViT-H/DC-AE at 512 pixels. Columns show the initial marked output, the raw generator output after 500 clean updates, and the returned output after quality fallback. Rows are fixed source positions 0, 16, 32 and 48 from 64 final inputs, without outcome filtering. The middle column gives LPIPS and the historical four-metric preservation decision; a failure returns the source. Bit accuracy labels are rounded; owner detection uses strict accuracy $> 0.6875$. Thus a raw non-detection need not count as a preserved removal.

detector and preservation thresholds. Validation and final class sets are disjoint; their identities were used in earlier development studies, so this is not new-class confirmation. No owner scores enter checkpoint selection.

| Starting model | Updates | Anchor | Preserved | Removed | Joint removed |
|---|---|---|---|---|---|
| Original | 500 | 0.0 | 64/64 | 20/64 | 20/64 |
| Fixed | 500 | 0.0 | 64/64 | 21/64 | 21/64 |
| Adapted | 500 | 0.0 | 64/64 | 20/64 | 20/64 |

Table 15: Independent open-weight clean-training attacks against DistSeal UViT-H/DC-AE at 512 pixels. Original is the released marked generator; Fixed and Adapted are the two prediction-anchored meta-distilled variants. Updates counts clean attacker updates, and Anchor is its prediction-retention coefficient. Preserved counts returned outputs passing LPIPS $\leq 0.2$, SSIM $\geq 0.6$, image CLIP $\geq 0.9$ and class-alignment retention $\geq 0.95$. Removed counts returned images losing owner detection, before requiring preservation. Joint counts both outcomes. All fractions use 64 reserved final inputs per starting model, initially all detected at strict bit accuracy $> 0.6875$. Checkpoint selection uses 32 separate validation inputs without owner feedback. This experiment returns the marked source when the raw adapted image fails preservation. The reported counts concern returned outputs; the raw adapted generator can fail the pair rule. Updates and anchor coefficients are selected separately for each starting model.

The quality-fallback client achieves 20/64 joint removals for the released model, 21/64 for the fixed-reference model and 20/64 for the adapted-reference model. All 64 returned images pass preservation in each case. However, this client retains the supplied generator and returns its source image whenever the raw attack fails the gates. Before fallback, raw preservation passes are only 21/64, 23/64 and 21/64, and raw removals are 58/64, 56/64 and 57/64. Consequently, its preservation rate is a property of the two-path client, not evidence that the adapted checkpoint alone preserves every source image. The clean null has 0/64 detections, which remains too small for precise low-FPR validation.

Neither training variant reduces quality-preserving removal relative to the released model. The longer adaptation horizon therefore does not rescue this meta-distillation hypothesis. Following the rejection criterion frozen before training completion, this direction stops here and is not expanded to additional watermark families. A different image defense would require a distinct mechanism, a source-grounded novelty assessment and a new experiment. These negative results do not support a generally robust image watermark.

The counts above retain the original frozen quality rule. The post-hoc reanalysis applies a revised gate to the same saved pairs.

**Post-hoc image preservation analysis.** The revised rule targets preservation of the main objects and scene. Let $x$ be the marked source and $x'$ the raw attack output. Write $D_{\text{LPIPS}}$ for the perceptual distance, $C_{\text{img}}$ for the image-embedding cosine, and $R_{\text{cls}}$ for class-alignment retention as defined in Subsection 4.1. The post-hoc gate is

$$Q_{\text{rev}}(x, x') = \mathbf{1}\left\{D_{\text{LPIPS}}(x, x') \leq 0.2, \quad C_{\text{img}}(x, x') \geq 0.9, \quad R_{\text{cls}}(x, x') \geq 0.9\right\}, \tag{32}$$

with a failure assigned if any metric is nonfinite. We apply this rule to the saved pairs. Historical tables retain their original prespecified counts. The retained-rule tables use this revision, explicitly as post-hoc reanalysis, while the owner detection threshold remains unchanged. The model-assisted inspection used to inform the revision covered 32 validation pairs.

For the 500-step inner adaptation, joint removal rises from 20/64 to 29/64 for the released model. Both trained variants reach 28/64. Each has a revised joint removal rate one sample lower than its released-model baseline, a difference of $-1.56$ percentage points. The paired 95% bootstrap intervals

| Inner steps | Model | Historical joint | Retained joint | Candidate pass |
|---|---|---|---|---|
| 100 | original | 20/64 | 28/64 | 33/64 |
| 100 | fixed | 22/64 | 26/64 | 31/64 |
| 100 | adapted | 22/64 | 31/64 | 35/64 |
| 500 | original | 20/64 | 29/64 | 30/64 |
| 500 | fixed | 21/64 | 28/64 | 30/64 |
| 500 | adapted | 20/64 | 28/64 | 30/64 |

Table 16: Post-hoc preservation reanalysis of open-weight DistSeal UViT-H/DC-AE clean-training attacks, with 64 paired inputs per row. Inner steps gives the clean adaptation horizon used in training the defense candidates. Original is the released marked generator; Fixed and Adapted are the meta-distilled reference variants. Historical joint uses LPIPS $\leq 0.2$, SSIM $\geq 0.6$, image CLIP $\geq 0.9$ and class retention $\geq 0.95$. Retained joint uses LPIPS $\leq 0.2$, image CLIP $\geq 0.9$ and class retention $\geq 0.9$. Candidate pass counts raw adapted images passing the retained rule before any source fallback. Joint counts simultaneous preservation and loss of owner detection at bit accuracy $\leq 0.6875$. The two horizons have different generation seeds; comparisons are paired within each horizon. This reuses evaluated data.

for the differences are $[-7.81, 4.69]$ points for the fixed reference and $[-6.25, 3.13]$ for the adapted reference. Both include zero. The 100-step study uses different generation seeds, so each variant is compared with its own released-model baseline. These results reuse the saved attacks and provide no reliable evidence of a robustness gain.

Figure 10 shows why the removal count still depends on preservation. The detector misses the watermark after the attack, but the clock casing changes enough to fail the revised gate.




Figure 10: DistSeal UViT-H/DC-AE at 512 pixels: left is the released marked source at final position 30, right its raw output after the selected open-weight clean-training attack in the 500-step defense study. This pair was selected post hoc as the highest-LPIPS owner non-detection among the 64 original-model candidates. LPIPS 0.329, image CLIP 0.753 and class retention 0.816 all fail the retained limits 0.2, 0.9 and 0.9; the label also gives historical SSIM 0.586. The clock casing and scene structure change, so this removal is not a joint success.

# 6 TextSeal 2 under probability access

## 6.1 Probability interfaces and rewriting attacks

All clients in this section use a locally simulated closed-source probability interface. They can submit prefixes and consume advertised token probabilities, but cannot read or update target weights. The independent reference model is public. The output-only reference-rewriting control in the pruning table is labelled separately. These contracts differ from the open-weight image attacks in the preceding section.

Let $\mathcal{V}$ be the vocabulary of size $V \in \mathbb{N}_{>0}$, $h \in \mathcal{V}^*$ a token prefix, and $k \in \mathcal{K}$ a fixed owner key in the scheme's key space $\mathcal{K}$. The vector $p(\cdot \mid h) \in \Delta(\mathcal{V})$ gives the underlying model probabilities after sampling filters. For next token $Y \in \mathcal{V}$, the key-conditional sampler law is

$$q_k(v \mid h) = \Pr(Y = v \mid h, k) \in [0, 1], \qquad q_k(\cdot \mid h) \in \Delta(\mathcal{V}). \tag{33}$$

The probability averages over the sampler's fresh noise at the fixed key. In the ideal fresh-context construction, averaging over independent keyed randomness restores the underlying marginal, whereas the fixed-key law carries the signal.

A client sampling exactly from $q_k(\cdot \mid h)$ at each prefix reproduces the keyed token law for that prompt. For example, the underlying model might offer two tokens with equal probability while a particular key favors one. Returning the underlying probabilities lets a client undo that preference through resampling. Returning the conditional probabilities retains it under exact resampling. Rewriting a marked answer adds another dependency because the answer itself appears in the prompt and can carry key evidence.

The implemented interface estimates each token's conditional winning probability in the clock race. Rather than count winners alone, it averages each token's chance of winning conditional on simulated competitor clocks. The raw estimate is $\hat{q} = (\hat{q}_v)_{v \in \mathcal{V}} \in [0, 1]^V$. Given positive normalizing sum $s_q = \sum_{v \in \mathcal{V}} \hat{q}_v \in (0, \infty)$, the advertised normalized estimate is $\tilde{q} = \hat{q}/s_q \in \Delta(\mathcal{V})$. Normalization introduces finite-draw bias. The interface exposes at most ten positive-probability alternatives; truncation and client reweighting can further change the law.

For probability vectors $a, b \in \Delta(\mathcal{V})$, total variation distance is $\mathrm{TV}(a, b) = \frac{1}{2} \sum_{v \in \mathcal{V}} |a_v - b_v| \in [0, 1]$. It equals the greatest discrepancy in probability assigned to any token subset. Two audits compare the 256-draw estimate with high-budget race frequencies. The mean and maximum observed distances are $0.00079$ and $0.02363$ on 16 calibration prefixes, and $0.00431$ and $0.03535$ on 32 saved attack prefixes. The second set spans 1 to 170 active tokens. Native clipping and the ideal race agree on the winner in every coupled reference draw in both audits. The reference frequencies have sampling error, and these checks cover only the tested prefixes. Full tables are in `tex/generated/probability_estimation_audit.tex` and `tex/generated/probability_attack_prefix_audit.tex`.

**Probability-exposure pilot: a copying control** The interface candidate in Section 6.1 is first evaluated on 32 newly reserved HC3 questions. Each source answer is generated by the native TextSeal2 implementation. Four blind rewriting arms compare pre-watermark versus estimated key-conditional probability exposure, each with direct resampling or confidence-dependent reference smoothing. Every arm produces three attempts per source; all 384 attempts are saved before quality evaluation or owner scoring. The quality rule requires Jina similarity at least $0.9$, bidirectional NLI entailment without truncation, exact preservation of the extracted number set, finite reference perplexity at most $1.5$ times the original, and no empty, capped or degenerate output. The first passing attempt is selected without detector feedback.

The selected outputs preserve almost all source token-context pairs: mean descriptive four-token-pair overlap is $0.9927$, $0.9221$, $0.9636$ and $0.9930$ in table order. These overlaps use saved generation token IDs and are not calibrated hypothesis tests. Literal string equality would misleadingly report zero exact copies because punctuation and formatting change. The observed two versus zero removals

| Exposed law / attack | Preserved | Joint removals | Word copies |
|---|---|---|---|
| Pre-watermark / direct | 32/32 | 0/32 | 30/32 |
| Pre-watermark / smoothing | 32/32 | 2/32 | 25/32 |
| Conditional / direct | 32/32 | 0/32 | 26/32 |
| Conditional / smoothing | 32/32 | 0/32 | 29/32 |

Table 17: Copying-control pilot for TextSeal 2 on Qwen3.5-4B instruct: 32 HC3 source answers, all detected at native $p \leq 0.001$. Every client has simulated probability-only target access. Pre-watermark exposes underlying probabilities; Conditional exposes the estimated keyed law. Direct resamples advertised tokens; smoothing mixes independent Qwen3.5-0.8B probabilities. Preserved counts outputs passing the frozen semantic, numeric, NLI, completion and perplexity-ratio rule; Joint removals counts preserved owner non-detections. Selection takes the first quality-passing attempt among three. Word copies means equality after Unicode normalization, case folding and word extraction; the pilot mostly copies marked answers.

under smoothing cannot establish a meaningful defense benefit when the baseline attack mostly copies marked text. A subsequent development pilot strengthens the paraphrasing instruction and forbids source four-token completions in both target and reference supports, with explicit accounting when no allowed probability mass remains. It retains the same quality gates. This iteration reuses the 32 sources, so it is development rather than independent confirmation.

**Why a clean-oracle argument is insufficient here.** Even a pre-watermark model distribution is evaluated on a prompt containing the original marked answer. That answer depends on the secret key. Therefore the rewrite can preserve key information through its input, although the probability computation itself does not consult the key. An impossibility argument for key-independent prompts and a clean resampling oracle does not apply to this experiment without accounting for that dependence. Conversely, the exact-law identity for conditional exposure does not guarantee robustness under smoothing, truncation or substantive rewriting. Both distinctions are required when interpreting the next pilot.

**Rewriting with source-pair blocking.** The completed second pilot keeps all 32 marked source answers and the same quality gates. Its stronger instruction and source four-token blocking substantially change the attack. The blocking operates after mixing target and reference supports; if no mass remains, the unmasked response is used and the fallback is recorded. This is an explicit exception rather than a guarantee of zero copied source pairs. The native detector is applied only after all attempts and the first quality-passing selections are frozen.

This test shows a development benefit: valid removals decrease from 25 to 5 for direct resampling and from 22 to 12 for smoothing. Preservation passes change from 25 to 24 and from 22 to 27, respectively, so the lower joint removal is not explained solely by rejecting more conditional-interface outputs. Nevertheless, this is a single-key, 32-source development comparison with three attempts per arm. The source identities have already been used in the copying control. The fresh human-null diagnostic remains 0/32 at the native nominal threshold, which cannot precisely validate a $0.1\%$ false-positive rate. The following experiments evaluate adaptive ratio attacks and fresh prompts. Additional keys and stronger conditional-estimator checks remain outstanding.

**Adaptive probability reweighting defeats conditional exposure** The source-blocked pilot in Section 6.1 motivates a stronger client rather than a robustness claim. On the same 32 development sources, an independent Qwen3.5-0.8B reference supplies probabilities on the target API's advertised top-ten token IDs. At a fixed prefix, let $\mathcal{V}$ be the vocabulary and $C \subseteq \mathcal{V}$ its nonempty advertised positive-probability support, of size $K_C = |C| \in \{1, \ldots, 10\}$. For token $v \in C$, the scalar advertised probability is $q_v \in (0, 1]$ and the numerically floored reference probability is $r_v \in (0, 1]$. The vectors

| Exposed law / attack | Access | Preserved | Removed | Joint removed | Word copies |
|---|---|---|---|---|---|
| Pre / direct | P | 25/32 | 32/32 | 25/32 | 0/32 |
| Pre / smoothing | P | 22/32 | 32/32 | 22/32 | 0/32 |
| Conditional / direct | P | 24/32 | 7/32 | 5/32 | 0/32 |
| Conditional / smoothing | P | 27/32 | 16/32 | 12/32 | 0/32 |

Table 18: TextSeal 2 simulated probability-only development experiment with Qwen3.5-4B instruct on 32 HC3 source answers, all originally detected at native $p \leq 0.001$. Pre uses underlying probabilities; Conditional uses the estimated keyed sampler law. Access: P is simulated probability-only target access; O uses only the saved marked answer. The independent reference is Qwen3.5-0.8B. Preserved counts first-quality-passing selections under Jina cosine $\geq 0.9$, bidirectional untruncated entailment, exact number sets, finite perplexity no greater than 1.5 times the original, and completion/repetition checks. Removed counts selected outputs evading owner detection regardless of quality; Joint counts preservation and removal together. Word copies counts normalized source equality. All fractions retain the 32-input denominator. Three attempts are frozen before quality selection and owner scoring. These sources are reused development data.

$(q_v)_{v\in C}, (r_v)_{v\in C} \in (0,1]^{K_C}$ contain the advertised-support coordinates before client normalization. With scalar tilt strength $\beta \in [1,\infty)$, define weights $w_v = q_v^{1-\beta} r_v^{\beta} \in (0,\infty)$ and normalized client law

$$a_v = \frac{w_v}{\sum_{u\in C} w_u} \in (0,1], \qquad (a_v)_{v\in C} \in \Delta(C). \tag{34}$$

The three attempts use $\beta \in \{1,2,4\}$. At $\beta = 1$, advertised probabilities cancel and the client samples the reference distribution restricted to $C$. Increasing $\beta$ favors tokens that the reference rates more highly relative to the advertised model. The implementation excludes zero-probability lanes, computes weights in log space with an explicit numerical floor, and applies the same source-pair blocking after reweighting. This is an exposed-support adaptation of prior ratio attacks, not a replication of their repeated-query protocol or an optimality result (Omidi et al., 2026).

| Exposed law / attack | Access | Preserved | Removed | Joint removed | Word copies |
|---|---|---|---|---|---|
| Pre / ratio tilt | P | 27/32 | 32/32 | 27/32 | 0/32 |
| Conditional / ratio tilt | P | 21/32 | 32/32 | 21/32 | 0/32 |

Table 19: TextSeal 2 simulated probability-only development experiment with Qwen3.5-4B instruct on 32 HC3 source answers, all originally detected at native $p \leq 0.001$. Pre uses underlying probabilities; Conditional uses the estimated keyed sampler law. Access: P is simulated probability-only target access; O uses only the saved marked answer. The independent reference is Qwen3.5-0.8B. Preserved counts first-quality-passing selections under Jina cosine $\geq 0.9$, bidirectional untruncated entailment, exact number sets, finite perplexity no greater than 1.5 times the original, and completion/repetition checks. Removed counts selected outputs evading owner detection regardless of quality; Joint counts preservation and removal together. Word copies counts normalized source equality. All fractions retain the 32-input denominator. Three attempts are frozen before quality selection and owner scoring. These sources are reused development data.

Every quality-selected output evades the native detector under both interfaces. Quality passes are 27/32 for the pre-watermark interface and 21/32 for conditional exposure. Consequently, the smaller conditional joint-removal count is entirely quality attrition: it is not evidence that the watermark survives. This negative control overturns a general robustness interpretation of the earlier direct-resampling result. All attempts precede owner evaluation, and selection still uses the first attempt passing the unchanged quality gates.

The 128 fresh-source confirmation cohort was frozen before this development outcome. It evaluates all six unguarded arms and therefore tests whether both the restricted benefit and adaptive failure transfer to new questions. It does not test the later support-pruning candidate, does not supply independent keys, and must not be relabeled as confirmation of that candidate.

## 6.2 Confirmation on fresh questions

The confirmation protocol reserves 128 new HC3 question identities, excludes the 1,712 retained questions and all 96 initial development/calibration identities, and splits generation into two disjoint 64-question shards. The shards share 64 separate public calibration identities. They use the same frozen watermark key and six source-blocked clients: direct resampling, reference smoothing and surrogate-ratio reweighting under each probability interface. Every client produces three attempts per source before quality-only selection and owner evaluation. The 2,304 attempts are retained. This cohort was fixed before the ratio development result and contains no support-pruning treatment.

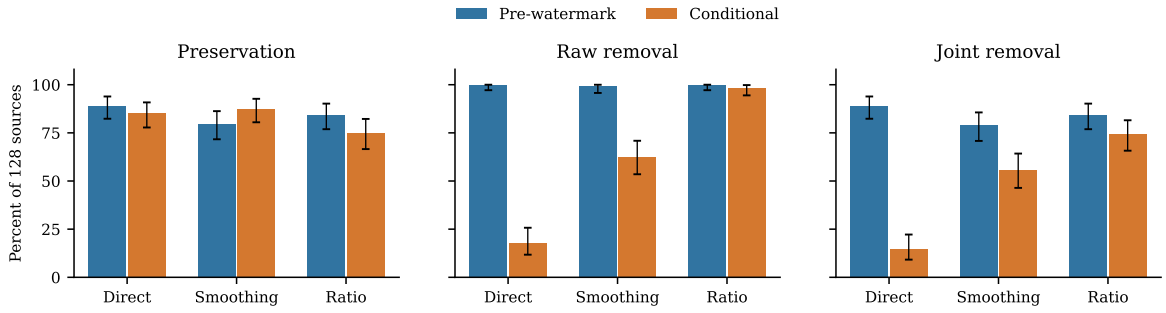



Figure 11: Fresh-source TextSeal 2 confirmation on Qwen3.5-4B instruct: 128 HC3 answers, one frozen key, all sources detected at $p \leq 0.001$. Each group names the probability-only client (direct resampling, confidence smoothing or reference-ratio reweighting); colors distinguish pre-watermark from estimated key-conditional probability exposure. Panels report preservation, owner non-detection and their joint event, each over all 128 sources after quality-only selection among three attempts. The auxiliary reference is Qwen3.5-0.8B. Error bars are exact binomial 95% intervals conditional on these models and key; joint removal alone cannot separate retained signal from quality attrition.

| Interface / client | Preserved | Removed | Joint removed | Joint 95% CI |
|---|---|---|---|---|
| Pre / direct | 114/128 | 128/128 | 114/128 | [82.3, 93.9]% |
| Pre / smooth | 102/128 | 127/128 | 101/128 | [70.8, 85.6]% |
| Conditional / direct | 109/128 | 23/128 | 19/128 | [9.2, 22.2]% |
| Conditional / smooth | 112/128 | 80/128 | 71/128 | [46.4, 64.3]% |
| Pre / ratio | 108/128 | 128/128 | 108/128 | [76.9, 90.2]% |
| Conditional / ratio | 96/128 | 126/128 | 95/128 | [65.7, 81.5]% |

Table 20: Fresh-source simulated probability-only confirmation for native TextSeal 2 on Qwen3.5-4B instruct, 128 HC3 questions and one frozen watermark key. All 128 source answers are detected at $p \leq 0.001$. Pre exposes underlying pre-watermark probabilities; Conditional exposes their estimated keyed sampler law. Direct resamples, smooth mixes an independent Qwen3.5-0.8B reference, and ratio reweights advertised tokens using that reference. Preserved counts outputs passing the frozen text rule; Removed counts owner non-detections regardless of preservation; Joint counts both outcomes. Every removal fraction uses the 128 originally detected sources. Selection takes the first quality-passing attempt among three, before owner scoring. Intervals are exact binomial 95% intervals conditional on these models and key.

All 128 originals are detected. Figure 11 compares preservation, raw removal and joint removal for each client. Under direct resampling, changing exposure reduces joint removal from 114/128 to 19/128, while preservation passes change from 114/128 to 109/128. The paired difference is $-74.2$ percentage points, with a descriptive 95% paired prompt-bootstrap interval $[-82.0, -65.6]$. Under smoothing, joint removal decreases from 101/128 to 71/128, with a paired difference of $-23.4$ points and interval $[-32.8, -14.1]$. These results confirm a restricted-client benefit on fresh questions.

The adaptive client changes the interpretation. Ratio reweighting removes 128/128 watermarks under pre-watermark exposure and 126/128 under conditional exposure. The latter preserves 96/128 outputs, of which 95 evade detection. Its 95/128 joint successes therefore reflect only one detected quality-preserving rewrite. The paired joint difference against pre-watermark exposure is $-10.2$ points, with interval $[-18.0, -2.3]$, but quality attrition explains most of that gap. A statistically distinguishable joint-success difference is not equivalent to robust watermark retention.

The intervals resample paired prompt identities 10,000 times with a fixed analysis seed. They are descriptive post-hoc uncertainty summaries, not prespecified multiple-comparison tests or uncertainty over training seeds and keys. The pooled human-null diagnostic is 0/128 at the native nominal threshold $0.001$; this small single-key set cannot validate a $0.1\%$ fixed-key false-positive rate precisely. No selected rewrite is a normalized word copy. Token-context overlap, source-blocking fallback counts and both shard-level tables remain in the analysis receipts.

### 6.3 Support pruning

A client can reweight the advertised tokens, including those carrying weak owner evidence. Support pruning removes some alternatives while retaining most of their estimated probability mass.

**The optimization at one prefix.** Let $\mathcal{V}$ be the vocabulary of size $V \in \mathbb{N}_{>0}$, $h \in \mathcal{V}^*$ a prefix, and $k \in \mathcal{K}$ an owner key in the scheme's key space $\mathcal{K}$. At that prefix, $\tilde{q} = (\tilde{q}_v)_{v\in\mathcal{V}} \in \Delta(\mathcal{V})$ is the normalized estimate of the keyed sampler law. The ordinary advertised support is a nonempty set $C_0 \subseteq \mathcal{V}$ of at most ten positive-probability tokens. The native owner evidence function is $e_k : \mathcal{V}^* \times \mathcal{V} \to [0, \infty)$. For token $v \in C_0$, it returns scalar $e_k(h, v) \in [0, \infty)$. Larger evidence favors detection. Given scalar mass-loss budget $\delta \in [0, 1)$, choose a nonempty subset $S \subseteq C_0$ solving

$$\max_{\emptyset\neq S\subseteq C_0} \min_{v\in S} e_k(h, v) \quad \text{subject to} \quad \sum_{v\in S} \tilde{q}_v \geq (1-\delta) \sum_{v\in C_0} \tilde{q}_v. \tag{35}$$

The objective raises the weakest evidence a supported client can choose. Sort tokens by decreasing evidence and accumulate probability mass until the constraint holds, retaining ties at the cutoff. For probabilities $(.80, .15, .05) \in \Delta(\{1, 2, 3\})$ and evidence $(4, 3, .1) \in \mathbb{R}^3$, dropping the last token raises the minimum evidence from .1 to 3 while discarding only $5\%$ of mass.

**What the budget controls.** Let $b_0 = \sum_{v\in C_0} \tilde{q}_v \in (0, 1]$ and $b_S = \sum_{v\in S} \tilde{q}_v \in (0, b_0]$. Define the renormalized probability vectors $P_0, P_S \in \Delta(\mathcal{V})$ by

$$P_0(v) = \frac{\tilde{q}_v \mathbf{1}\{v \in C_0\}}{b_0}, \qquad P_S(v) = \frac{\tilde{q}_v \mathbf{1}\{v \in S\}}{b_S}. \tag{36}$$

On $S$, $P_S$ is $P_0$ scaled up by $b_0/b_S$. Outside $S$, $P_S$ is zero. Splitting the absolute differences gives

$$\mathrm{TV}(P_0, P_S) = \frac{1}{2}\left[\sum_{v\in S}(P_S(v) - P_0(v)) + \sum_{v\in C_0\setminus S} P_0(v)\right] \tag{37}$$

$$= \tfrac{1}{2}[(1 - b_S/b_0) + (1 - b_S/b_0)] = 1 - b_S/b_0 \leq \delta. \tag{38}$$

This bounds how much pruning changes the advertised distribution at this prefix. Sequential changes can accumulate, so answer preservation still needs direct evaluation.

For an output $y = (y_1, \dots, y_L) \in \mathcal{V}^L$ of length $L \in \mathbb{N}_{>0}$, let $\mathcal{T} \subseteq \{1, \dots, L\}$ contain the positions scored by the native first-occurrence rule. At each such position, the prefix is $h_t \in \mathcal{V}^*$, the retained support is nonempty $S_t \subseteq \mathcal{V}$, and its evidence floor is $a_t = \min_{v \in S_t} e_k(h_t, v) \in [0, \infty)$. If every scored token $y_t$ belongs to $S_t$, then

$$S_{\text{path}} = \sum_{t \in \mathcal{T}} e_k(h_t, y_t) \;\geq\; \sum_{t \in \mathcal{T}} a_t, \tag{39}$$

where $S_{\text{path}} \in [0, \infty)$ is the summed native evidence. A bound for saved token IDs applies to decoded text only when the scored path is preserved by retokenization. Reference smoothing and independent rewriting may choose tokens outside the retained supports.

**Support pruning resists a supported ratio client in development**

**A matched development comparison.** The experiment in Section 6.3 has completed generation, preservation selection and native owner scoring. It reuses the same 32 marked answers as the earlier blocked-source and ratio experiments. The source-answer files are byte-identical; the watermark key, sampling settings and quality-selection rule also match. Each interface uses its own public-prefix confidence calibration. The guard was designed after observing the earlier ratio failure, so this is a development comparison rather than an independent confirmation. Three attempts per source are frozen before selecting the first quality-passing output. No source is replaced by the original answer when all attempts fail.

| Exposed law / attack | Access | Preserved | Removed | Joint removed | Word copies |
|---|---|---|---|---|---|
| Prune 10 / direct | P | 23/32 | 4/32 | 3/32 | 0/32 |
| Prune 10 / smooth | P | 25/32 | 18/32 | 15/32 | 0/32 |
| Prune 10 / ratio | P | 25/32 | 4/32 | 3/32 | 0/32 |
| Prune 20 / direct | P | 23/32 | 3/32 | 3/32 | 0/32 |
| Prune 20 / smooth | P | 27/32 | 15/32 | 13/32 | 0/32 |
| Prune 20 / ratio | P | 24/32 | 2/32 | 1/32 | 0/32 |
| Independent reference | O | 22/32 | 32/32 | 22/32 | 0/32 |

Table 21: TextSeal 2 simulated probability-only development experiment with Qwen3.5-4B instruct on 32 HC3 source answers, all originally detected at native $p \leq 0.001$. Pre uses underlying probabilities; Conditional uses the estimated keyed sampler law. Access: P is simulated probability-only target access; O uses only the saved marked answer. The independent reference is Qwen3.5-0.8B. Preserved counts first-quality-passing selections under Jina cosine $\geq 0.9$, bidirectional untruncated entailment, exact number sets, finite perplexity no greater than 1.5 times the original, and completion/repetition checks. Removed counts selected outputs evading owner detection regardless of quality; Joint counts preservation and removal together. Word copies counts normalized source equality. All fractions retain the 32-input denominator. Three attempts are frozen before quality selection and owner scoring. These sources are reused development data. Prune 10 and Prune 20 retain at least 90% and 80% of the ordinary top-ten estimated probability mass. Independent reference rewrites outside the advertised support.

**A gain against supported reweighting.** Without pruning, conditional ratio reweighting removes all 32 selected watermarks, including all 21 quality-preserving rewrites. Retaining at least 90% of the ordinary top-ten estimated mass reduces raw removals to 4/32 and joint removals to 3/32, with 25/32 quality passes. At 80% retained mass, the corresponding counts are 2/32 raw removals, 1/32 joint

removals and 24/32 quality passes. Thus the development gain is not explained by rejecting more outputs. Among quality-preserving outputs, watermark removal falls from 21/21 to 3/25 and 1/24. All selected guarded outputs differ from their sources under normalized-word equality; this check excludes literal copying but does not certify substantive rewriting or factual equivalence.

**The access boundary is visible in the failures.** Direct conditional resampling has 5/32 joint removals without pruning and 3/32 at either guard budget, a small development difference. Smoothing does not improve: its 12/32 baseline joint removals become 15/32 and 13/32. The independent reference removes all 32 watermarks, including all 22 outputs passing the preservation gates. These controls show that supported ratio resistance does not establish resistance to external rewriting. They also prevent describing the guard as a generally robust watermark or a defense for open-weight models.

**Checking the mathematical relation against native scoring.** All 672 attempts were scored as actual decoded text and as saved generated token IDs. Every generated-ID statistic satisfies the frozen pathwise evidence bound, including the native first-occurrence masking rule. Decode–encode roundtrips hold for all 96 attempts in each direct and ratio arm, 96/96 and 94/96 in the two smoothing arms, and 92/96 independent-reference attempts. The remaining paths receive no decoded-text certificate. At the native nominal threshold, the bound certifies 40/96 ratio paths at 90% retention and 62/96 at 80% retention. These are certificates across all attempts, not quality-selected success counts; an uncertified path can still be detected. The relation does not prove the nominal Gamma calibration for adaptive paths. The separate human-null audit addresses one fixed-key diagnostic, rather than repairing this limitation.

**What remains to establish.** The existing 128-question confirmation concerns conditional exposure without support pruning. It cannot confirm this new treatment. Independent prompts, keys and model pairs, an attack that adapts to the guard, and additional watermark families are still required. The complete native scores, quality outcomes, support-path certificates and selected strings are retained in the hashed owner results and portable development receipts.

### 6.4 False-positive audit

The 128-source confirmation contains only 128 human-null prefixes, too few to assess a nominal $0.1\%$ error rate precisely. We therefore freeze a separate 4,096-prefix diagnostic before computing its owner scores. It uses HC3 human-labelled answers from question-hash buckets 65–89, outside the probability-interface evaluation and public-calibration buckets. All retained, development, confirmation and calibration question identities are excluded. Questions and first-100-word human prefixes are deduplicated, and known human-prefix hashes are excluded. No threshold or key is fitted on this set.

The native TextSeal2 v2 decision remains $p \leq 0.001$, with null-tail $p$-value $p \in [0, 1]$. It flags 5 of 4,096 prefixes: an empirical false-positive rate of $0.122\%$, with an exact binomial 95% interval $[0.040\%, 0.285\%]$. Every prefix has at least one scored context/token pair. This cohort is compatible with the nominal rate but does not precisely establish that rate for every key, dataset or adaptively selected input. The interval is descriptive under the usual independent-sample model, conditional on the existing key and this human-labelled population.

The audit runs entirely on CPU. It calls the released native copula-score primitive in batches, preserves the native first-occurrence context/token-pair rule, and uses the unchanged Gamma-tail function. Before computing the audit, it checks agreement with the scalar detector on 12 probes: empty, short and repeated-token cases plus eight fixed-position real prefixes. All scored-pair counts and per-token values agree to the declared $10^{-12}$ tolerance. The frozen manifest, individual $p$-values, source hash and result hashes remain in the audit receipt. This is an implementation and null-behavior check, not evidence of attack resistance or a new calibration procedure.

# 7 Conclusions and reproducibility

**What the evidence supports** The experiments show adaptation against fixed detectors and substantial suppression of several measured watermarks. They also expose limits of raw evasion: BIRA suppression accompanies lower bidirectional entailment; some DC-AE weight merges suppress detection while damaging class alignment; and probability resampling can bypass the sampler under pre-watermark probability access, which is a stronger oracle than access to only the marked conditional law.

None of these observations establishes universal watermark erasure. A negative result from one owner detector does not exclude an unknown second mark. Similarity metrics do not certify semantic equivalence or image usefulness. Preserved class alignment does not certify diversity, realism or absence of memorization.

**Reproducibility and evidence** The repository README gives the build and experiment commands. The detector catalogue and experimental records in `tex/records/` retain protocols, full tables and historical comparisons. The migration map in `tex/provenance/migration.json` identifies their original sources. The watermarking results index in `docs/watermarking/README.md` provides the fixed-sample galleries and complete example texts. Full recent rewriting examples are in `tex/generated/` and the corresponding sample JSON files are in `tex/provenance/`.

The latest experiment receipts in `tex/provenance/` include the frozen selections and owner scores.

- Conditional exposure: `text_exposure_confirmation_results.json`
- Support pruning: `text_exposure_guard_results.json`
- Human-null audit: `textseal2_human_null_audit.json`
- Retained 512-pair image audit: `retained_image_preservation_audit.json`
- Fixed qualitative examples and source hashes: `qualitative_evidence.json`
- Image preservation reanalysis: `revised_quality_no_ssim_results.json`

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